

Where we are

Assignment 1: grades released; questions → TA Taehyun

Assignment 2: due in two weeks (April 9)

Changes to syllabus: the last two programming labs → vibe coding

Final project: will be released this week

Proposal: due in three weeks

Discussion: in-class discussion

Milestone : in-class discussion

Final demo: May 14, final exam hours (can't be late)

Storytelling & Narrative

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

March 23, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

Using data to craft a
compelling **narrative**,
often involving **interactivity**

Value of storytelling

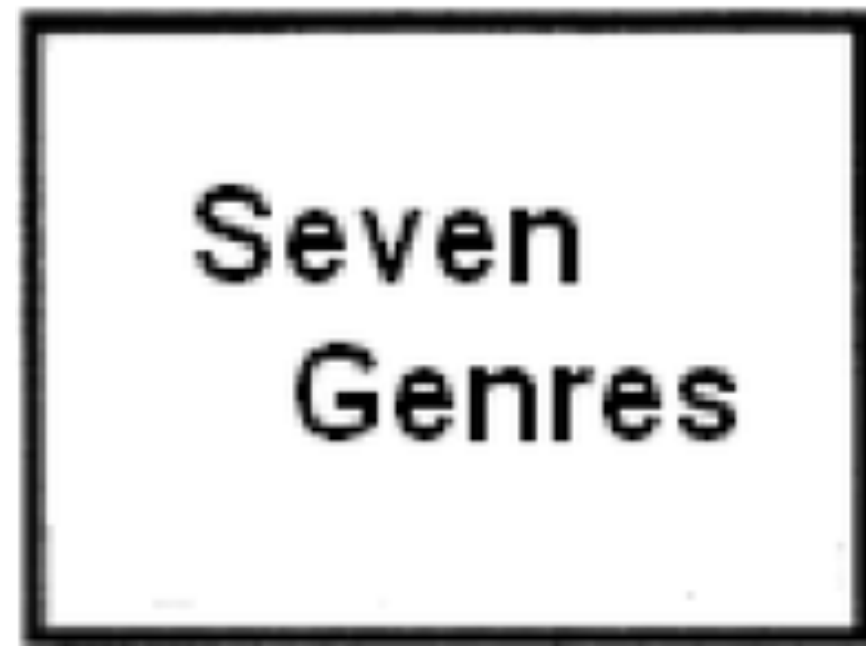
Get beyond **exploring** data to **engaging** and **explaining**



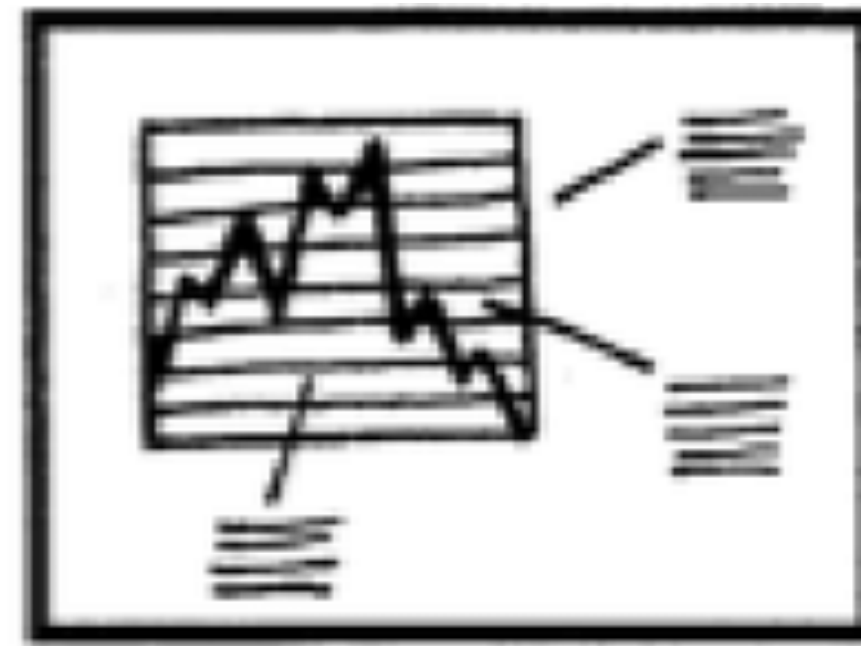
interactivity in storytelling
designer/reader balance

Engages through **conflict, resolution, uncertainty**

Narrative genres



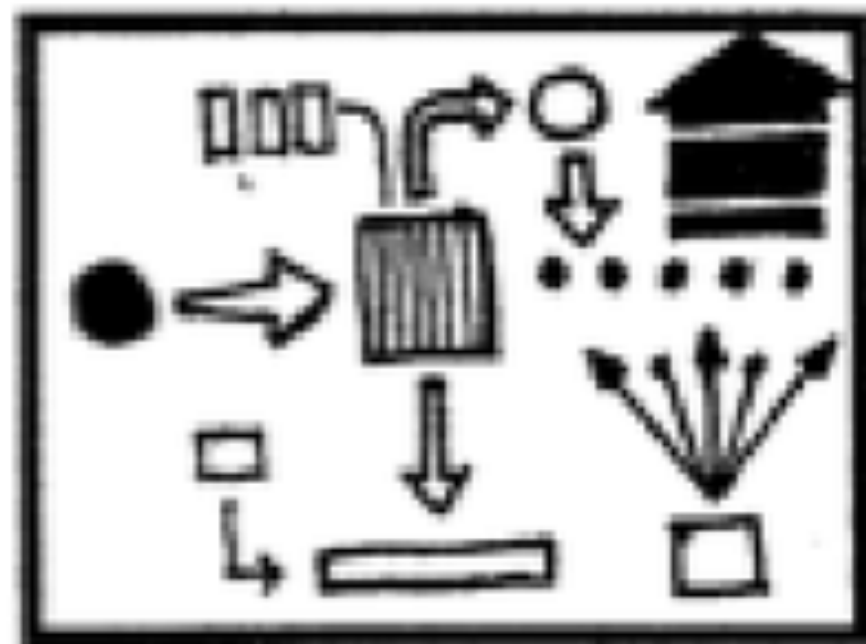
Magazine Style



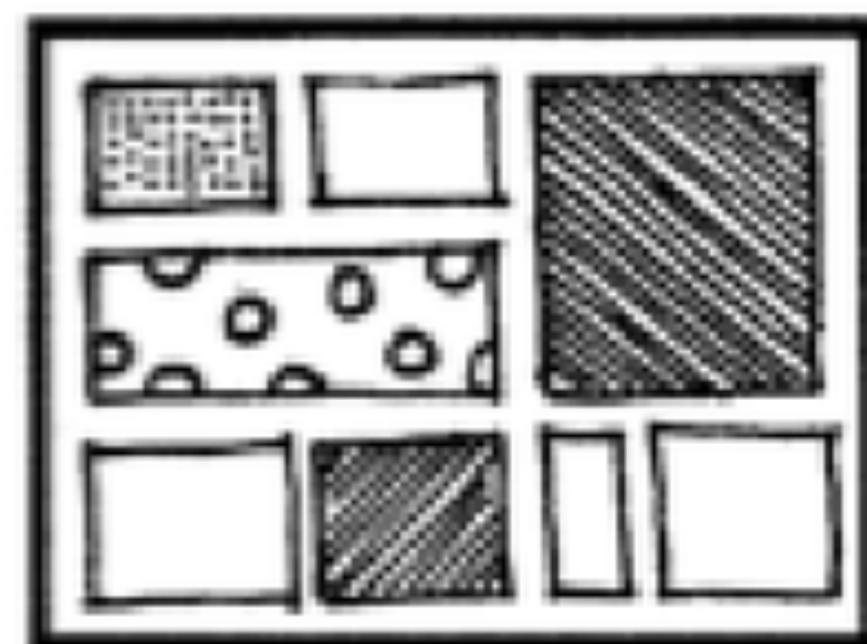
Annotated Chart



Partitioned Poster



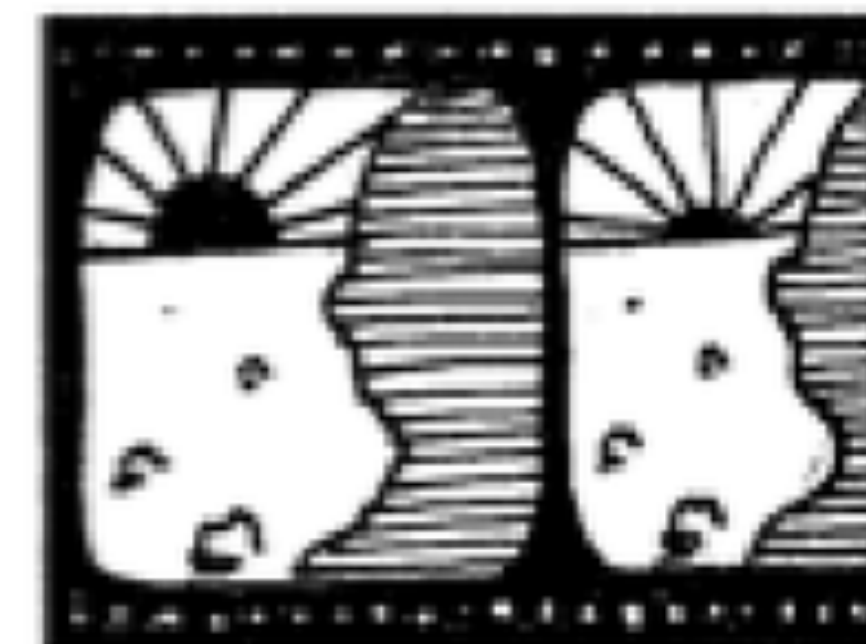
Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

Not mutually exclusive; differ in **ordering** and **designer/reader control**

Magazine style

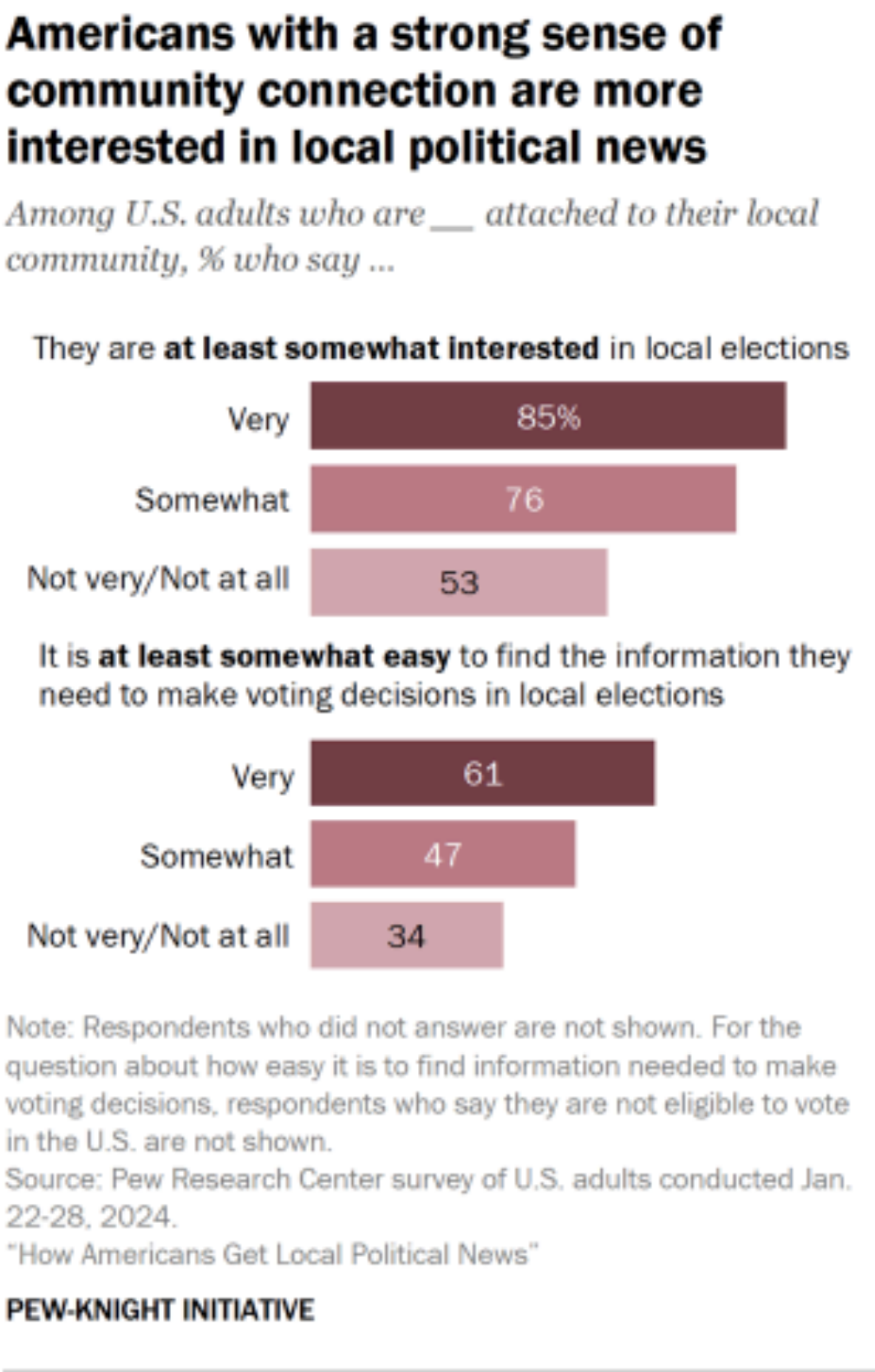
An image embedded
in a page of text



Americans who are more attached to their local community are more interested in and satisfied with local political news

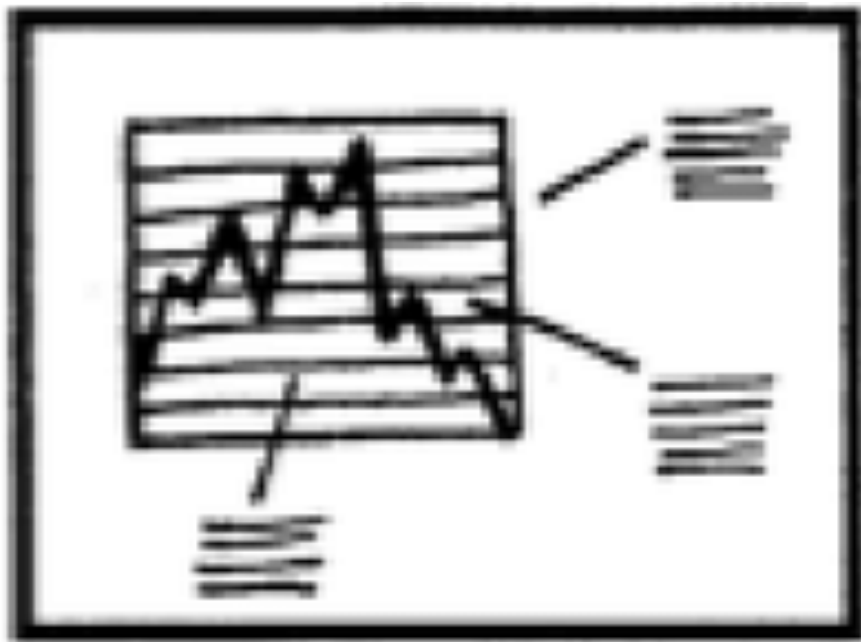
There are clear links between Americans’ sense of community attachment and their experiences with news on local government and politics. Those who feel more attached to their community have more interest in news about local laws and local elections, higher satisfaction with the quality of their local political news, and an easier time finding the information they need to vote.

- 91% of Americans who feel very attached to their community are at least somewhat interested in news about local laws and policies, compared with 68% of those who are not very or not at all attached. A similar pattern holds for interest in news about local elections.
- A majority of Americans with a strong sense of community connection (61%) say it is at least somewhat easy to find the information they need to make voting decisions in local elections. Among those with little or no sense of community attachment, just 34% feel this way.
- Among U.S. adults who follow news about local government and politics, four-in-ten of those with strong community attachment are extremely or very satisfied with the local political news they get – more than double the share among those who feel little or no attachment to their community (16%).



Annotated chart

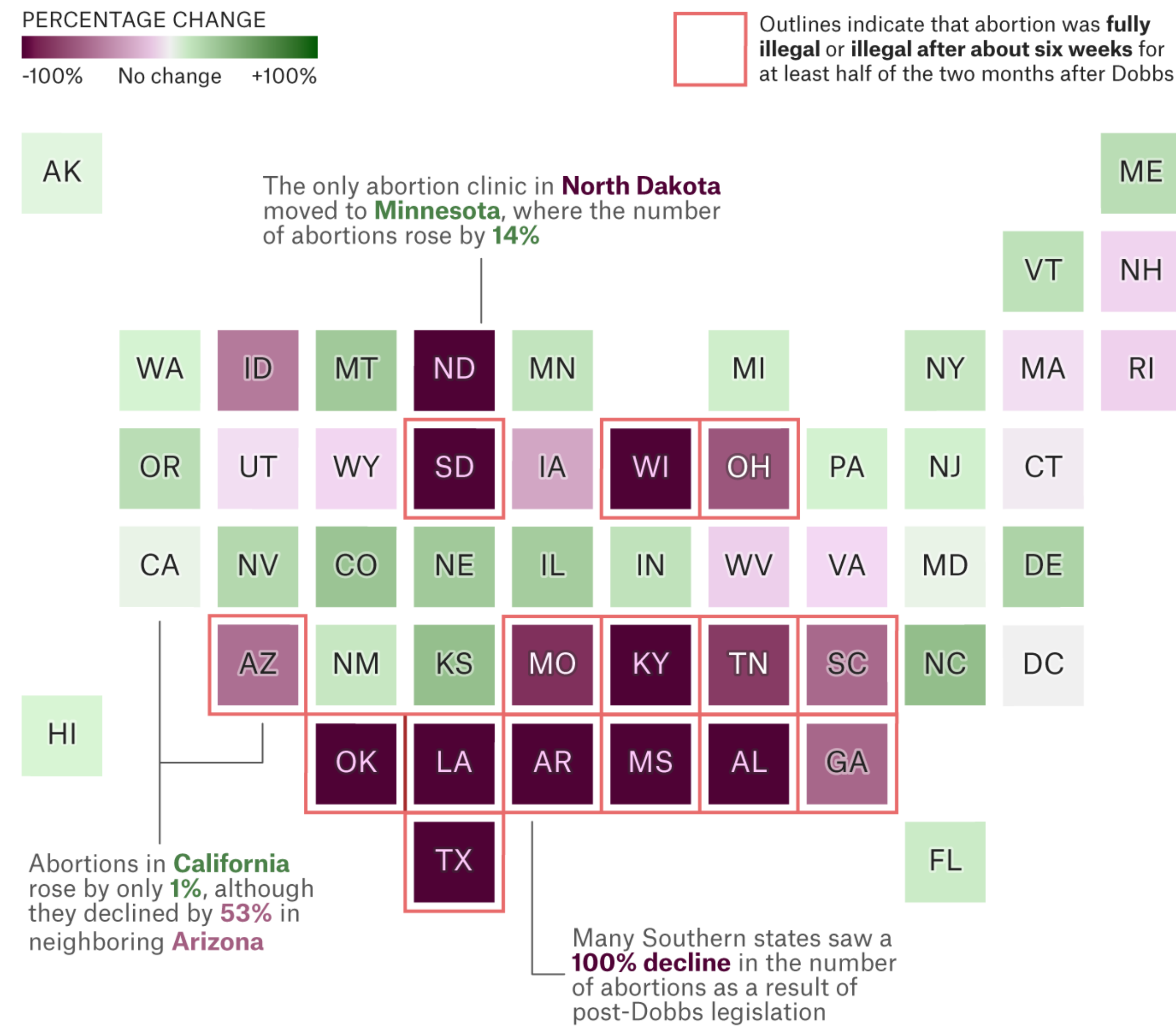
Usually with a catchy title
to tell the key point



The Dobbs v. Jackson Women's Health Organization ruling was a 2022 Supreme Court decision that overturned the constitutional right to abortion and eliminated federal standards on abortion access.

Several states saw a complete drop in abortions after Dobbs

Percentage change in number of abortions from April to August 2022, by state



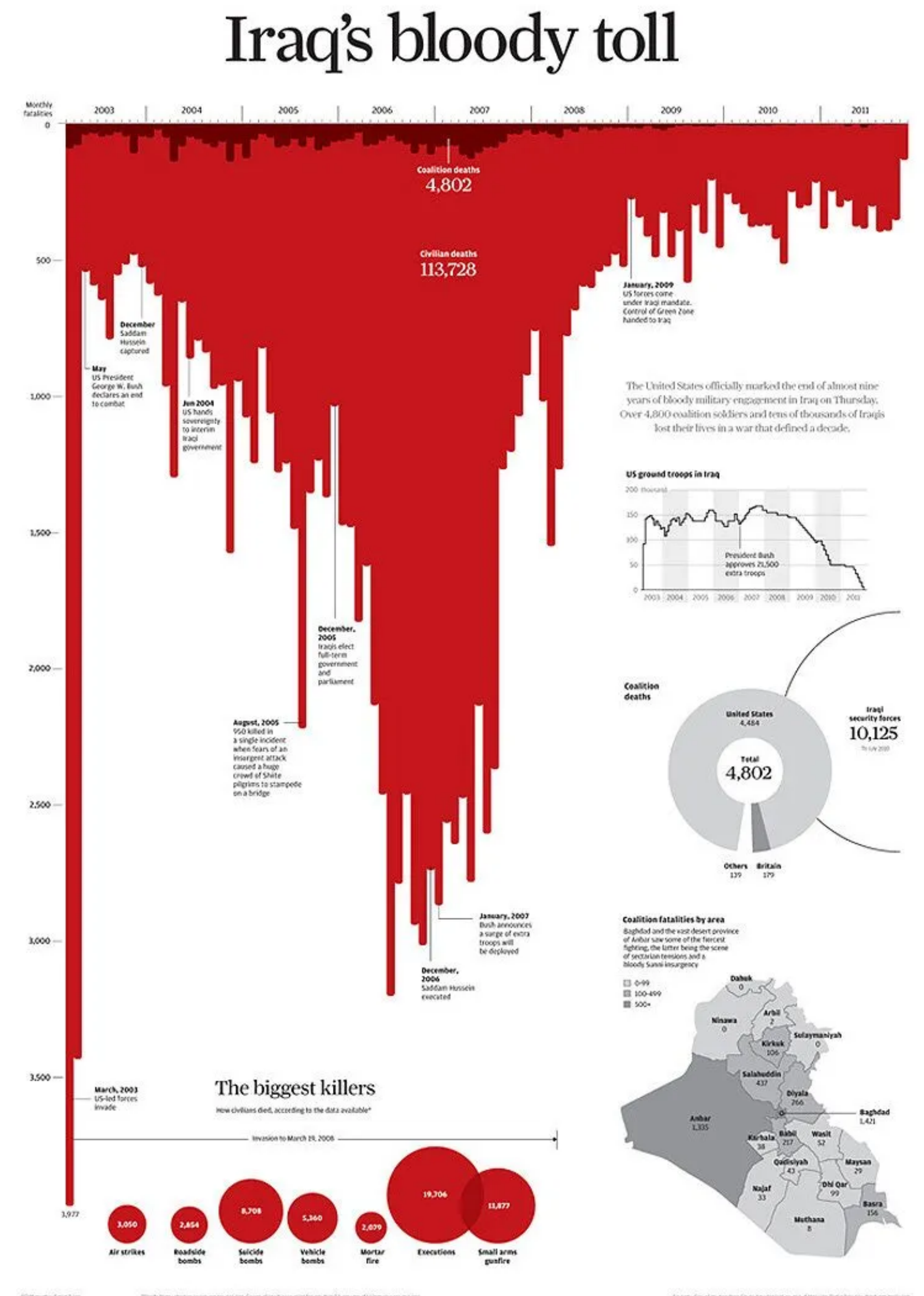
Percent change in abortions is calculated from #WeCount's unrounded data. Less-populated states might see larger percentage shifts because the raw number of abortions in those states tend to be smaller.

Partitioned poster

Usually with a catchy title
to tell the key point

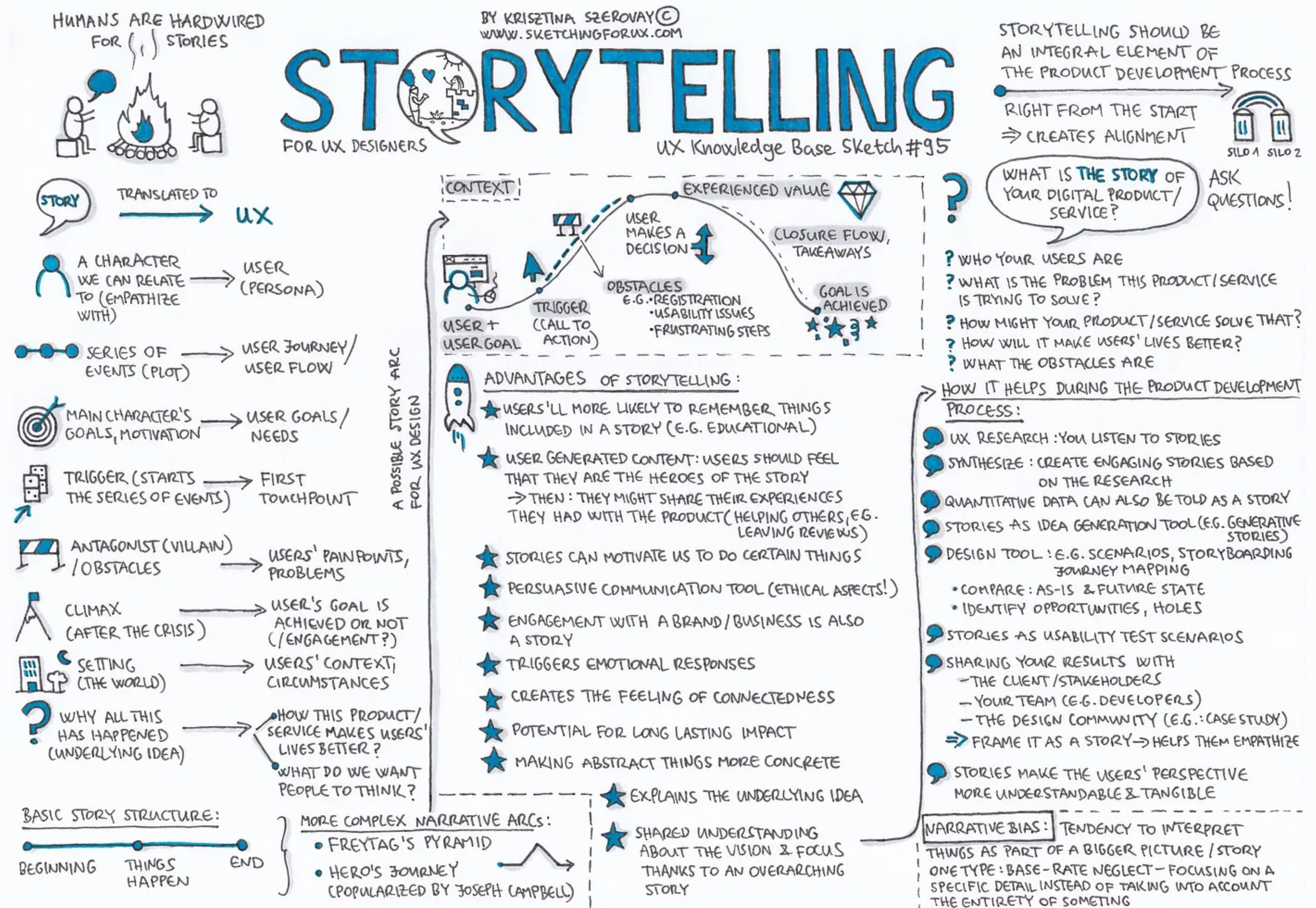
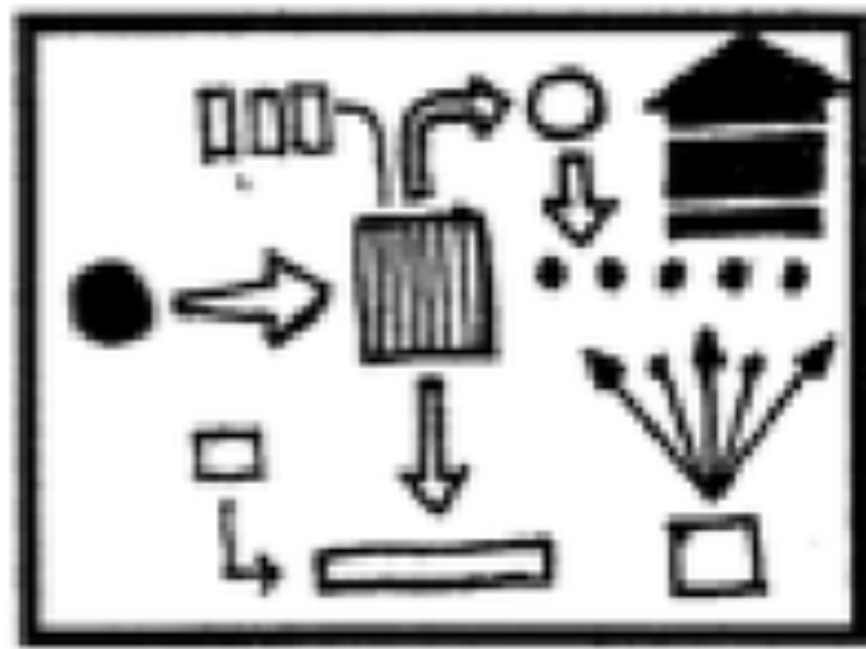


<https://www.simonscarr.com/iraqs-bloody-toll>

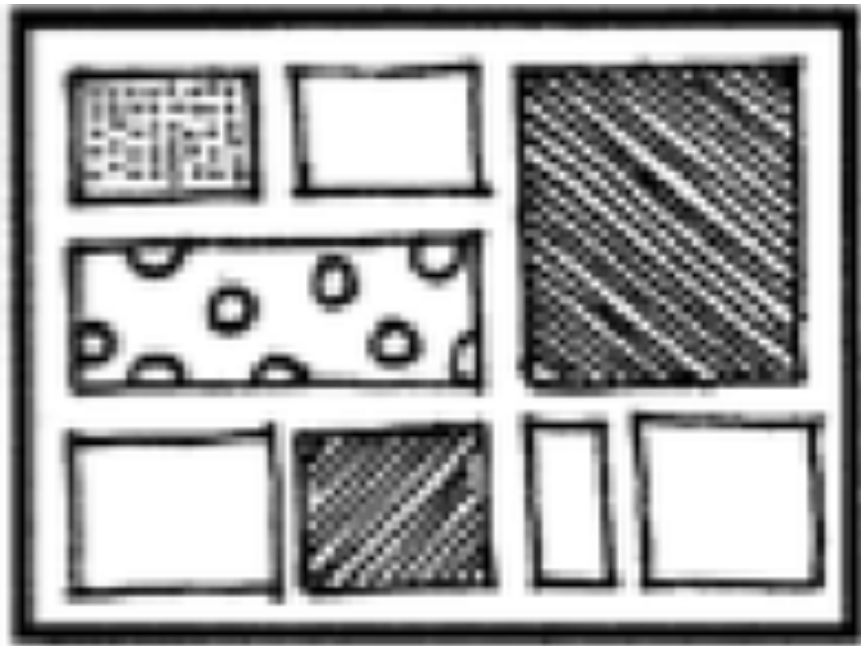


Flow chart

For lack of a better
example....



Comic strip



AN ILLUSTRATED GUIDE TO DATA LITERACY

PIE CHARTS

GOOD MORNING!

HEY BECKY!

BEFORE THE MEETING, I NEED TO SEE WHICH STATE IS THE BIGGEST PROPORTION OF OUR SALES.

SURE...

psst, BECKY... PROPORTION JUST MEANS A PART COMPARED TO THE WHOLE.

SURE...

MAYBE A PIE CHART? BUT AREN'T THEY BAD?

THEY SHOW PROPORTIONS WELL... REMEMBER ALL SLICES MUST ADD UP TO 100%.

OH NO, THAT SEEMS LIKE TOO MANY SLICES!

I WISH WE HAD TRIED A DIFFERENT CHART...

HOW'D IT GO?

SHE COULDN'T TELL WHICH SLICE WAS BIGGEST... SMALL DIFFERENCES ARE LOST IN A PIE CHART...

IT'D WORK BETTER FOR PROPORTIONS THAT ARE MORE RECOGNIZABLE: 25% 50% 75%

WAFFLES IN THE BREAK ROOM! THEY RAISED THE BAR TODAY!

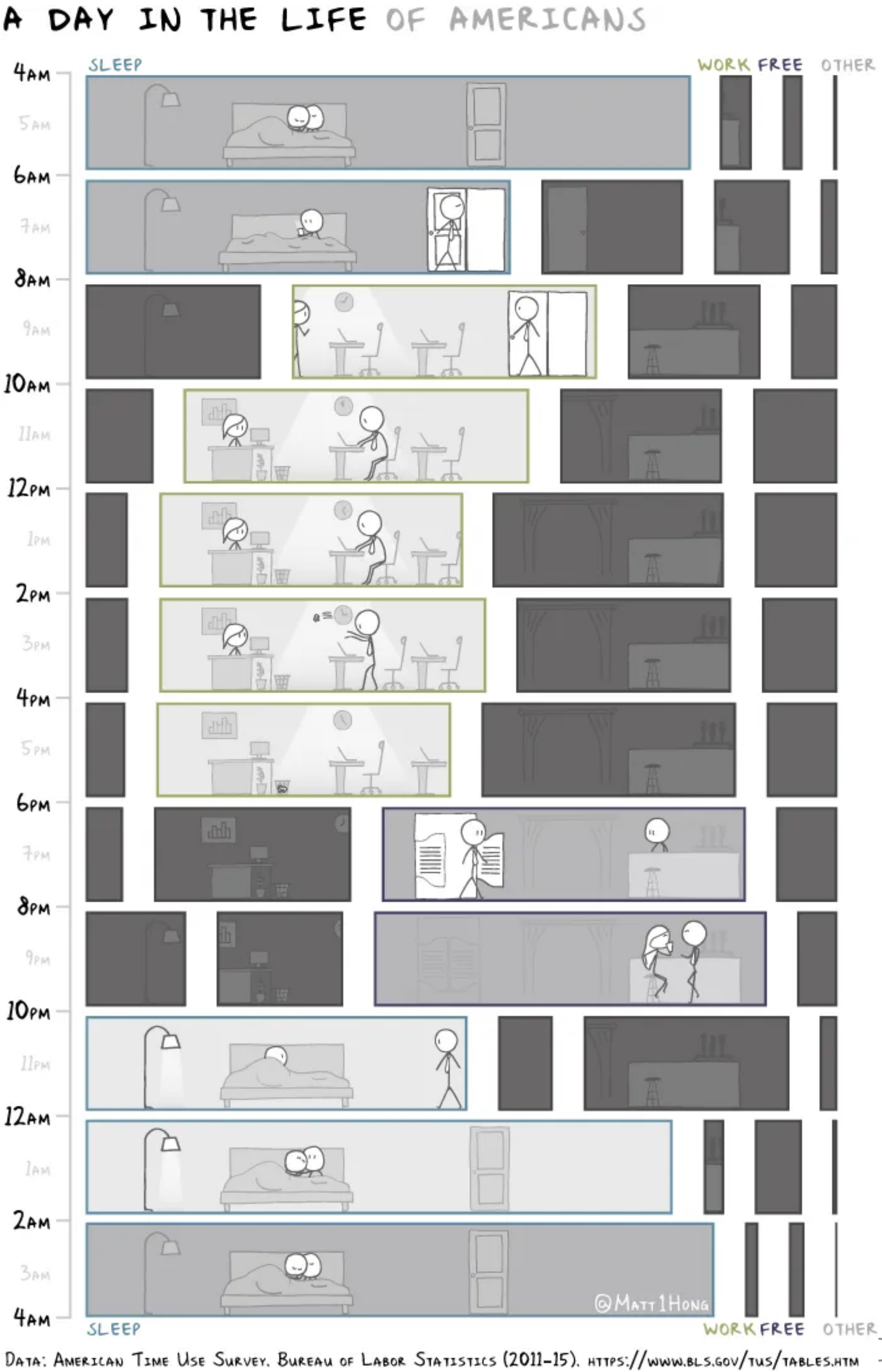
PROPORTION OF SALES BY STATE

20% 18% 15% 15% 13% 9% 5% 5%

VA GA MD NC SC FL NY NJ

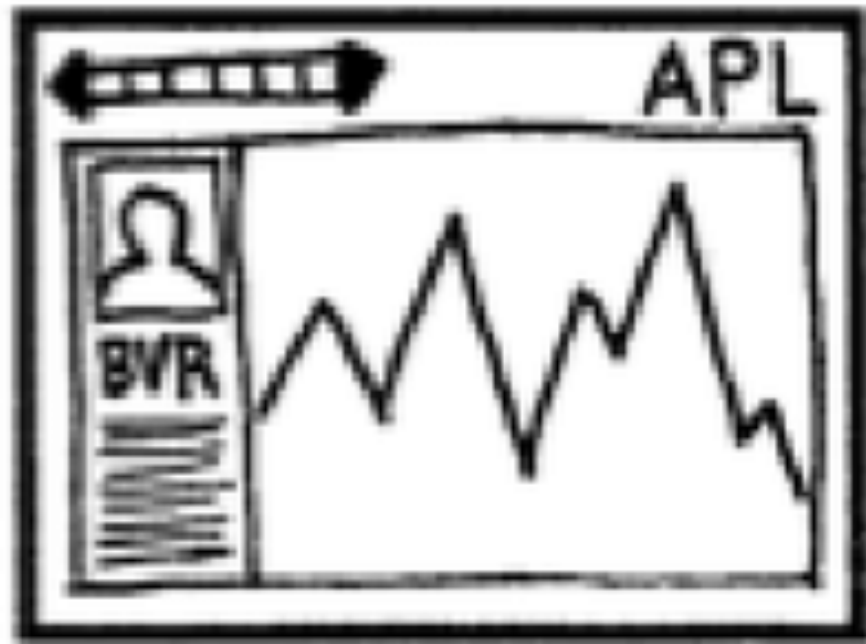
WE SEE THAT VIRGINIA IS THE BIGGEST PROPORTION OF SALES.

THE PIE CHART: A GOOD CHOICE WHEN YOU'RE SHOWING A PART-TO-WHOLE RELATIONSHIP WITH ONLY A FEW SLICES THAT HAVE LARGE DIFFERENCES. ALL SLICES SHOULD ADD UP TO 100%, OTHERWISE YOU MAY HAVE ADDED OR REMOVED A CATEGORY, OR IT'S NOT REALLY A PART-TO-WHOLE RELATIONSHIP. OTHER CHART TYPES TO TRY: BAR OR STACKED BAR CHART, TREEMAP, OR A WAFFLE CHART. NOT ALL PIE CHARTS ARE BAD; JUST BE THOUGHTFUL WITH YOUR CHOICES.



Slide show

Often with
interactivity



Scientific Proof that Americans are Completely Addicted to Trucks

By Adam Pearce Blacki Migliozi and David Ingold | Jan. 15, 2015

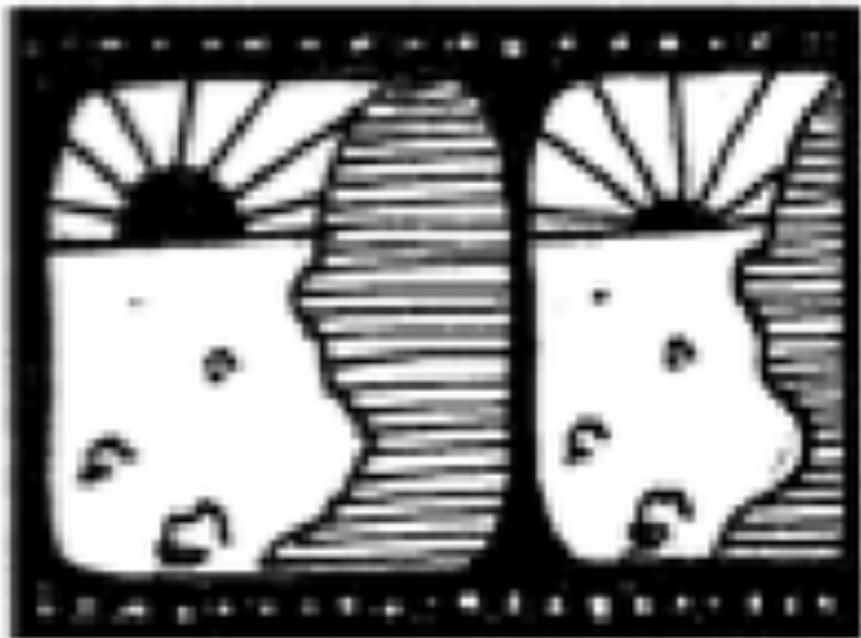
Car geeks will be elbowing each other out of the way for a glimpse of Ford's gorgeous new GT sports car when the Detroit auto show opens to the public this weekend. But when it comes to the new vehicles people actually buy, the story is all about light-trucks—the broad category that includes pickups, SUVs, crossovers, and minivans. As the economy gains strength and gas prices drop, Americans are turning away from cars and snapping up trucks of all sizes.

2014 new U.S. auto sales, by model

Scroll



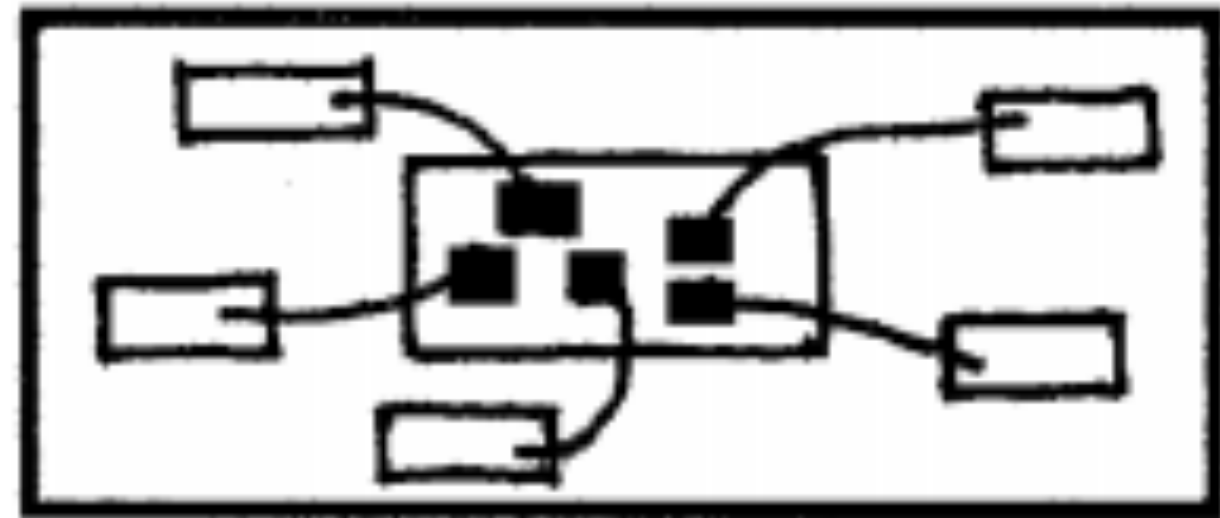
Film/Video



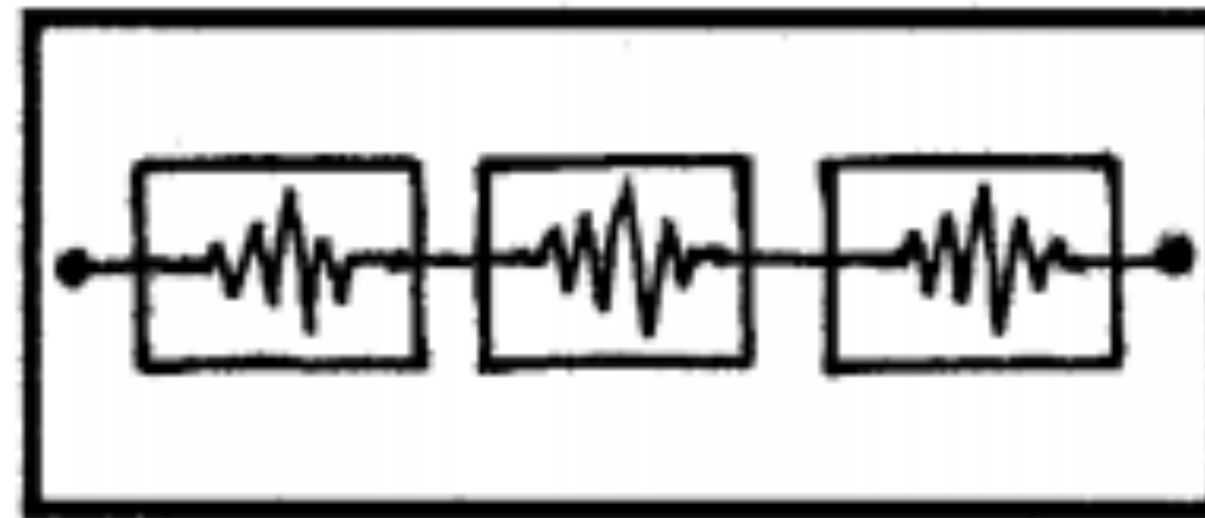
Slideshow might be the most
common and engaging format...

Techniques for storytelling: 3 types of interactivity

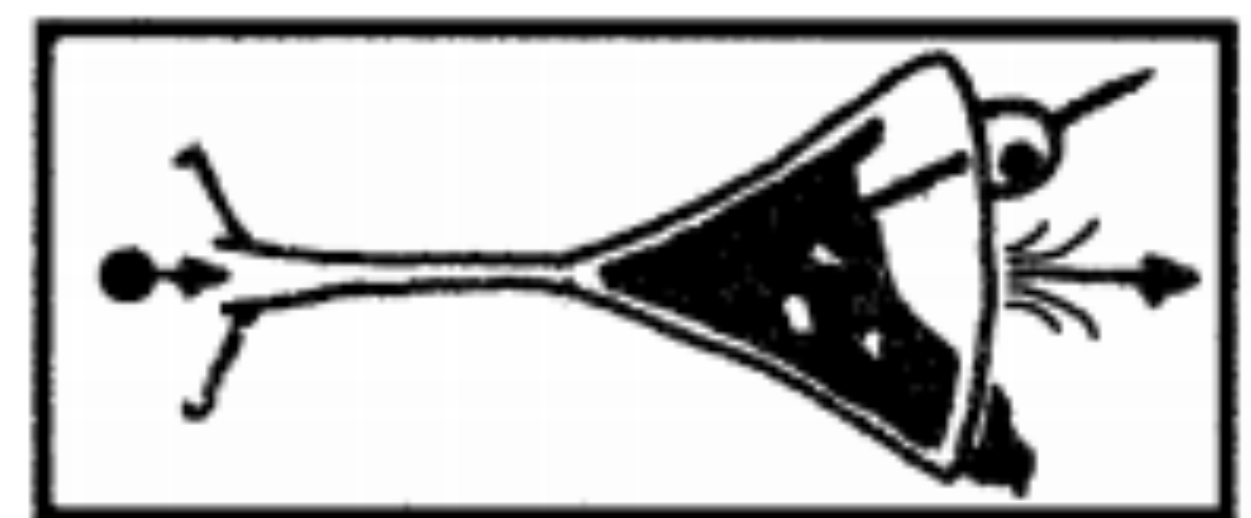
Drill-down story



Interactive slideshow



Martini glass



More
reader
control



More
designer
control

Some **inspirations** from movies...



https://www.youtube.com/watch?v=v4seDVfgwOg&ab_channel=EveryFrameaPainting



Montage / editing

Prescribed order

Where you look explicitly directed

Little reader/viewer agency

Ensemble staging / direction

Less prescribed order

Salience, structure nudges you to look

More reader/viewer agency

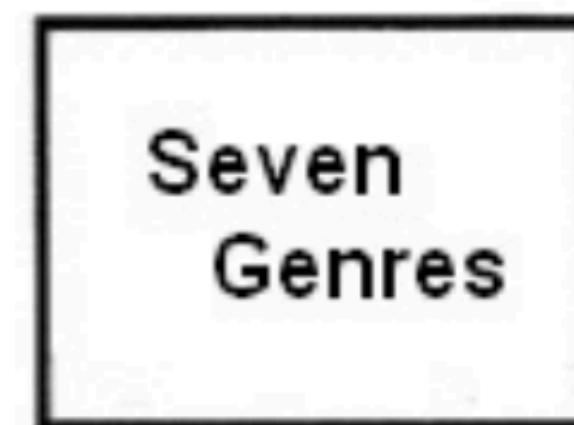
Film uses both, so can we! But the choice should be deliberate



Montage / editing



Ensemble staging / direction



Seven
Genres



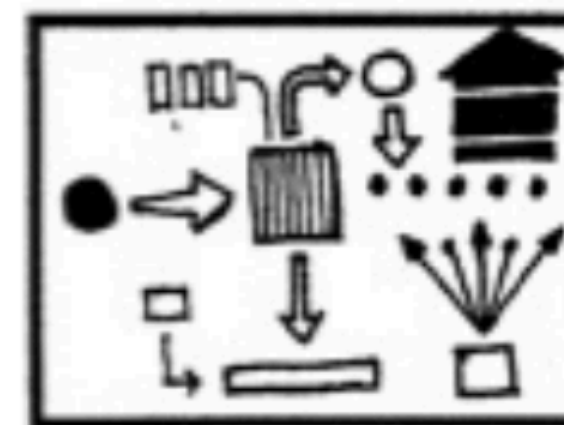
Magazine Style



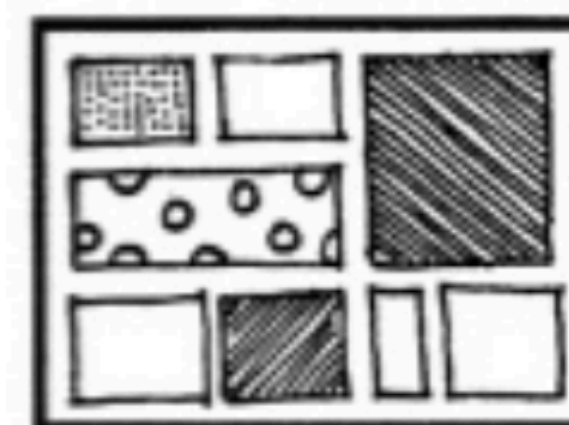
Annotated Chart



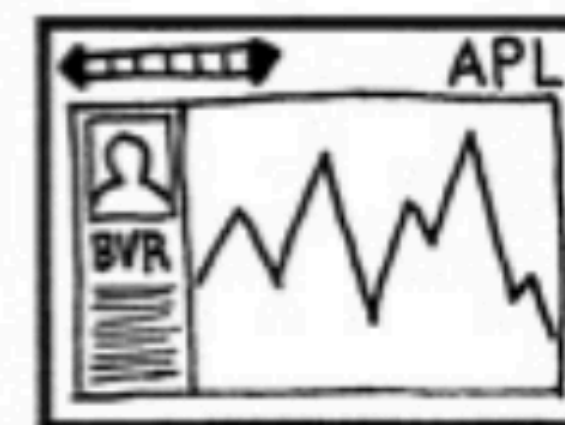
Partitioned Poster



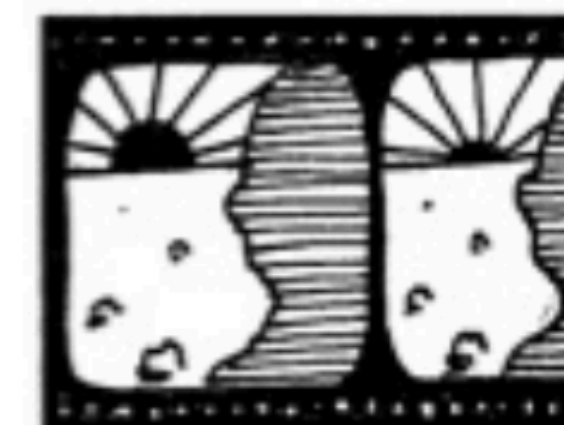
Flow Chart



Comic Strip



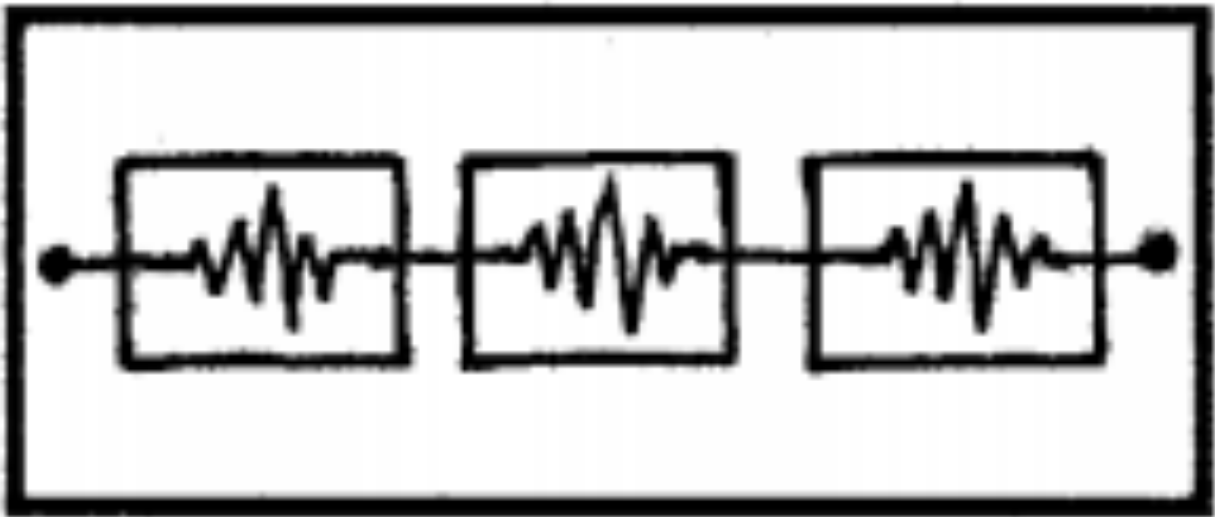
Slide Show



Film/Video/Animation



Montage / editing



Income Mobility Charts for Girls, Asian-Americans and Other Groups. Or Make Your Own.

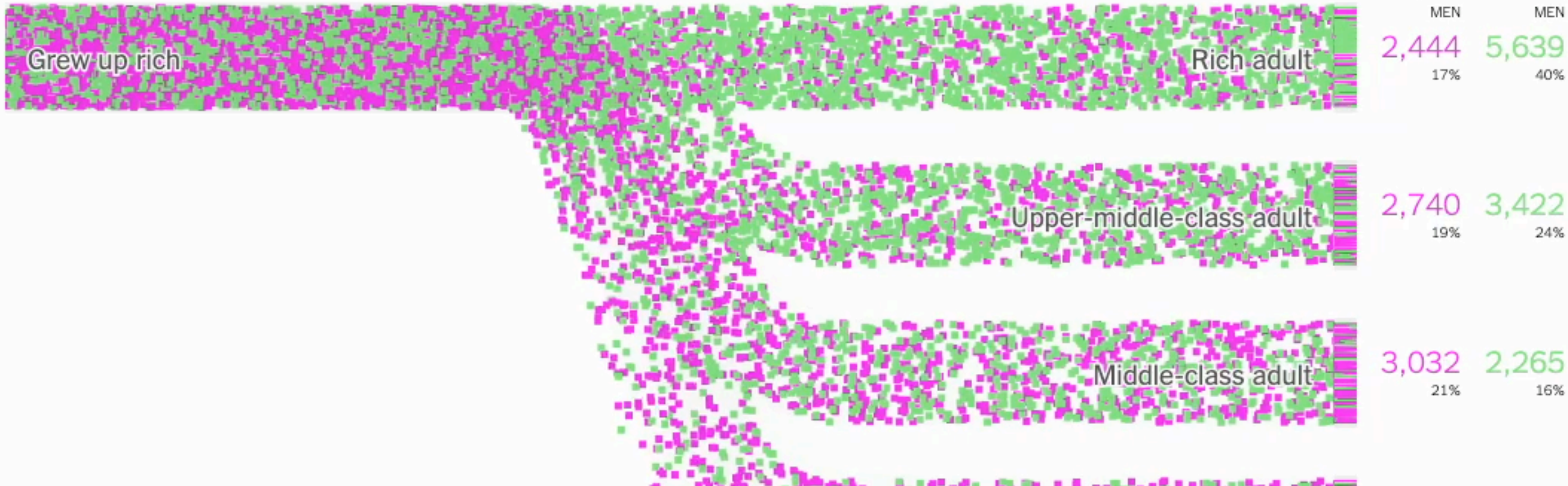
By EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE and KEVIN QUEALY MARCH 27, 2018

Last week we [wrote](#) about a sweeping new study of income inequality, which followed 20 million children in the United States and showed how their adult incomes varied by race and gender. The research was based on data about virtually all Americans now in their late 30s.

This first animated chart illustrates one of the study's main findings: White boys who grow up rich are likely to remain that way, while black boys raised in similarly wealthy households are more likely to fall to the bottom than stay at the top in their own adult households.

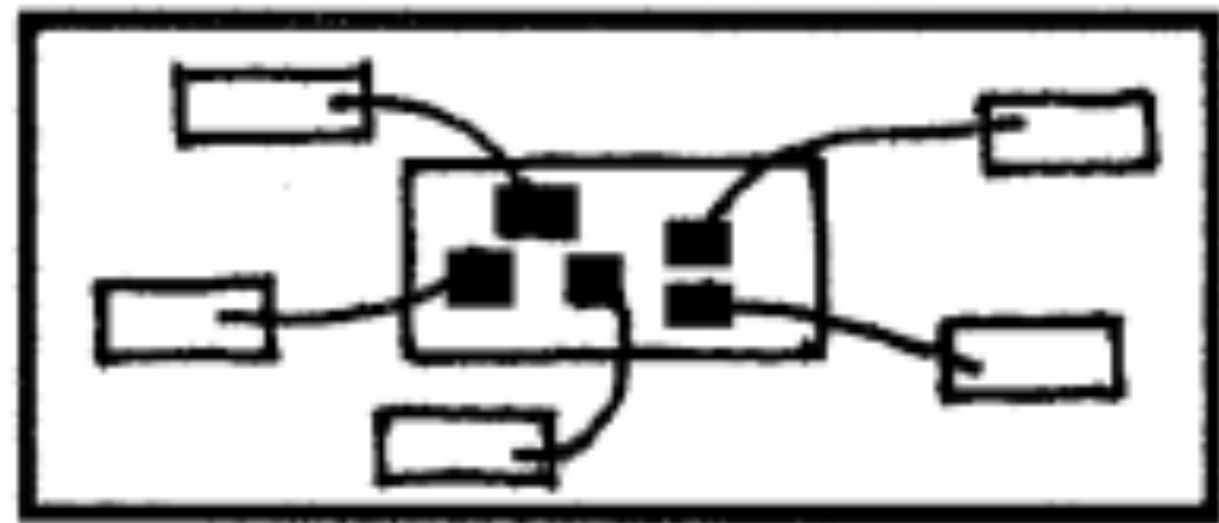
Black and **white** boys raised in wealthy families

Follow the lives of these **41,877** Americans and see where they end up as adults:



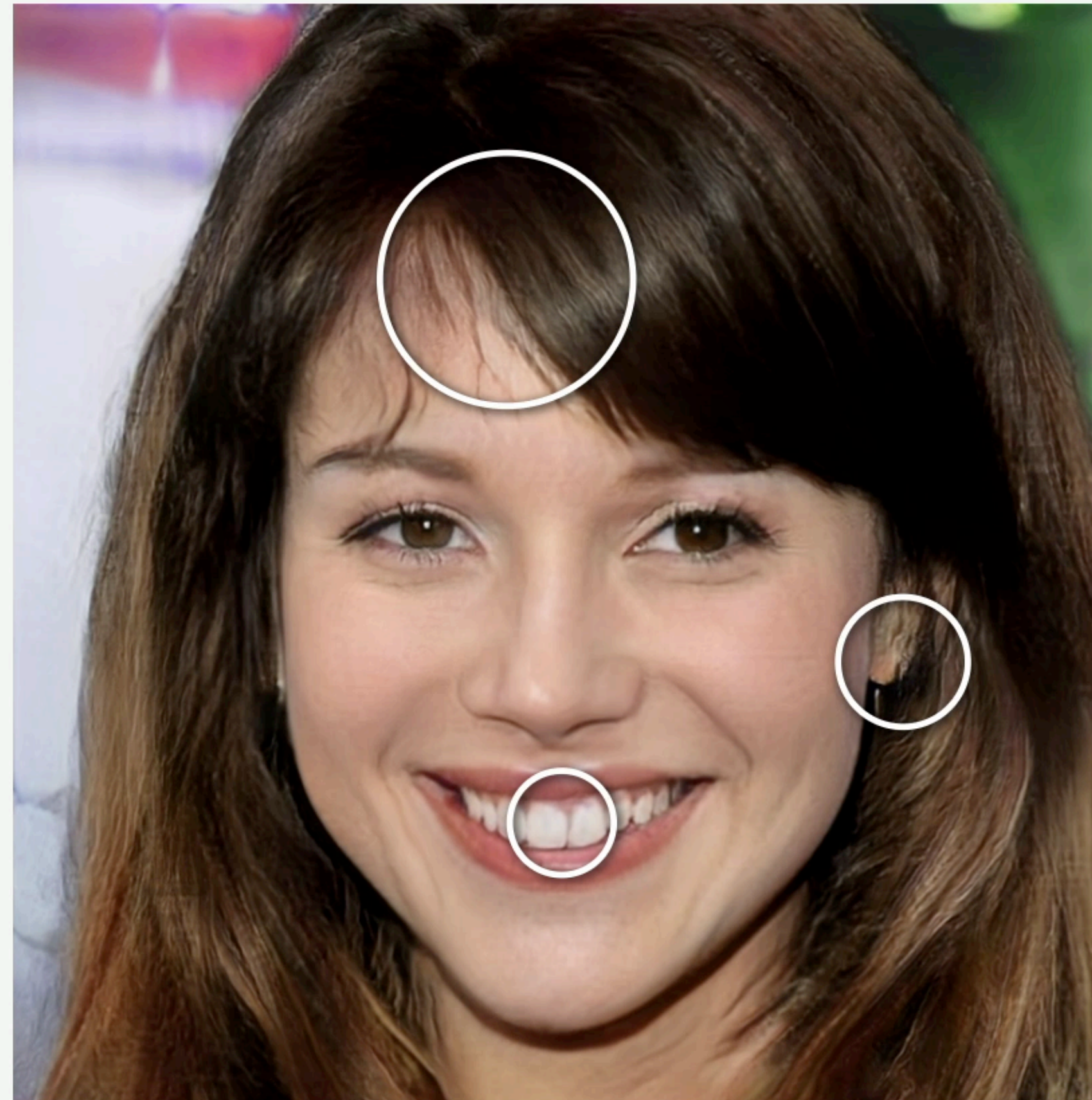


Ensemble staging / direction

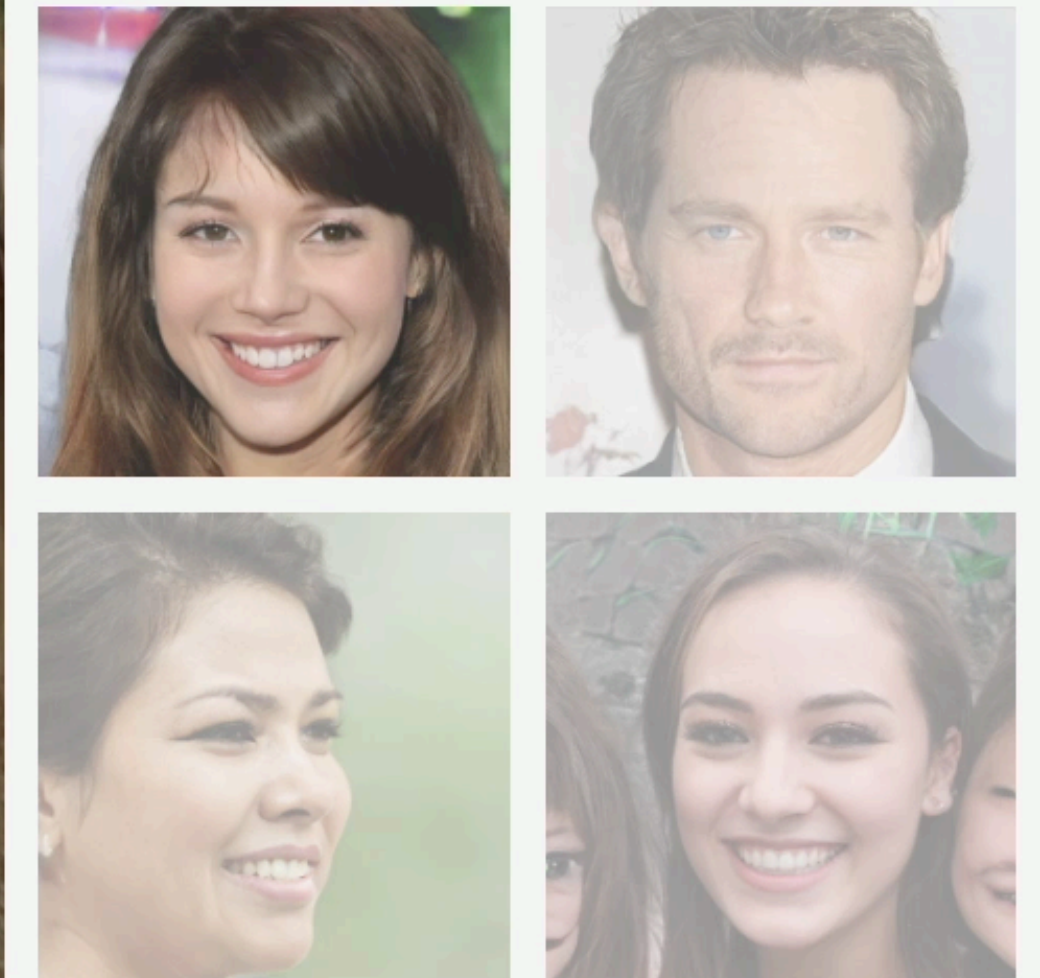


What gives away a machine-generated image?

Interactivity on illustrations can help people get more context around certain objects that may not have clear and separable boundaries.



Select region for more information.



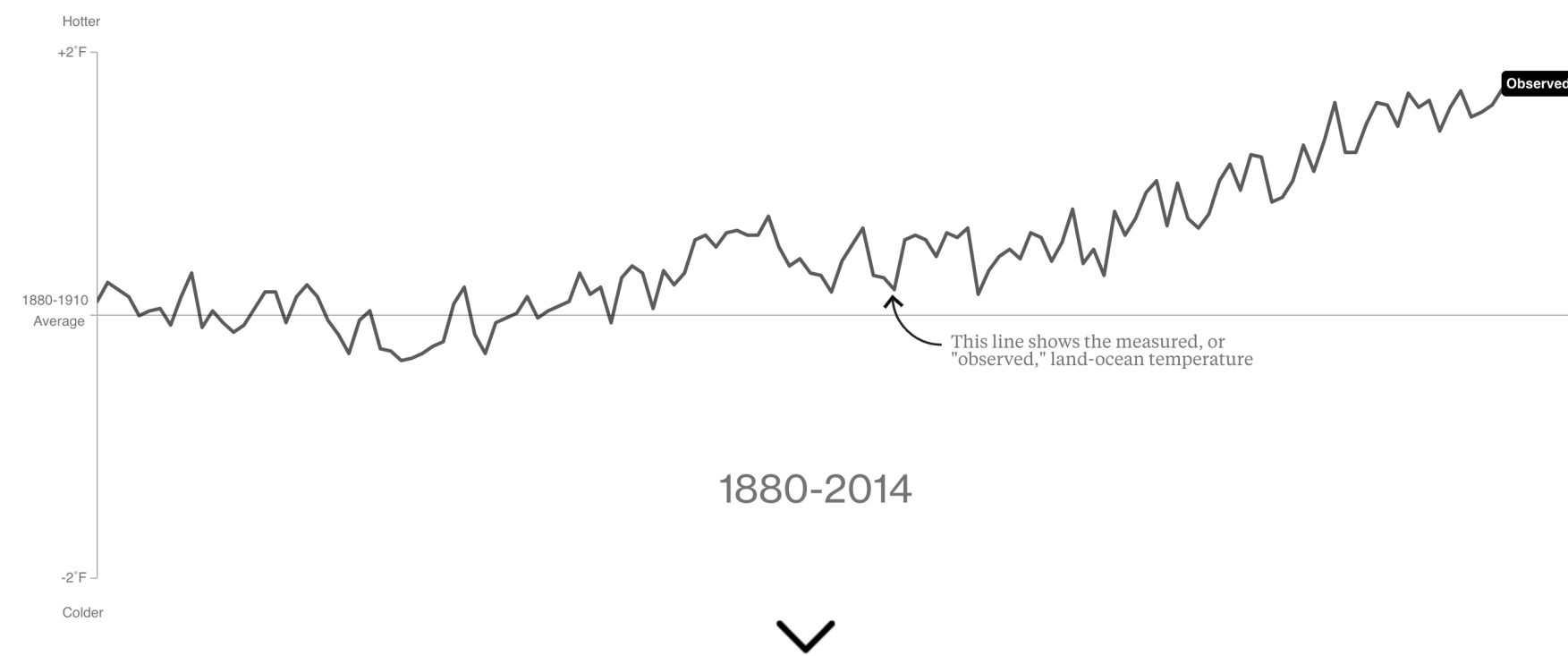
Best if you experience
them yourself...

Activity

What's Really Warming the World?

By Eric Roston and Blacki Migliozi | June 24, 2015

Skeptics of manmade climate change offer various natural causes to explain why the Earth has warmed 1.4 degrees Fahrenheit since 1880. But can these account for the planet's rising temperature? Scroll down to see how much different factors, both natural and industrial, contribute to global warming, based on findings from NASA's Goddard Institute for Space Studies.



<https://www.dropbox.com/scl/fi/lp7uctpq86adav6gid6p2/Screen-Recording-2024-11-14-at-8.25.41-PM.mov?rlkey=0n8hjk2yy37nqnbzd8xikz0&dl=0>

What techniques?

How are they directing attention?

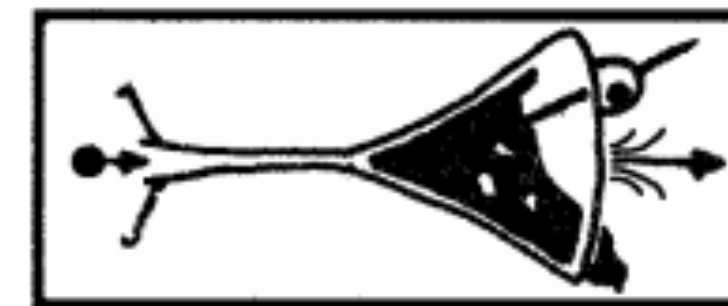
Is it effective / engaging?



Montage / editing



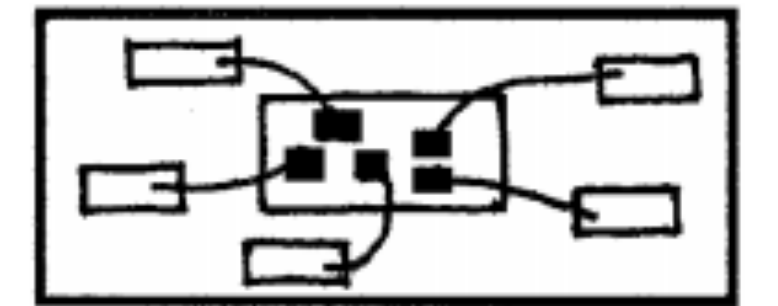
Ensemble staging / direction



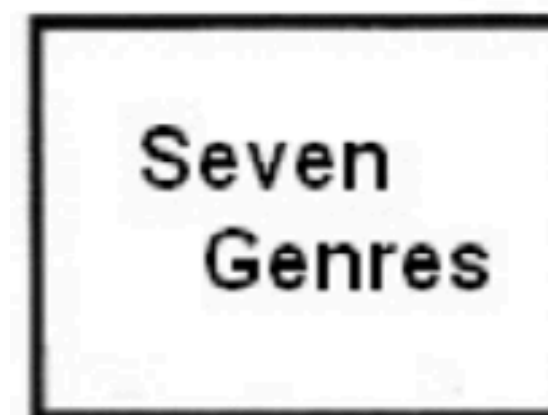
Martini glass



Interactive slideshow



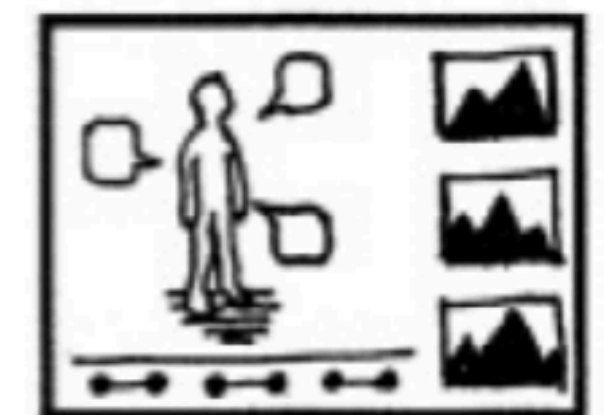
Drill-down story



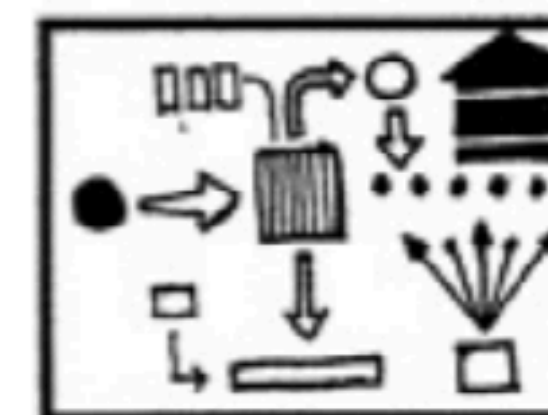
Magazine Style



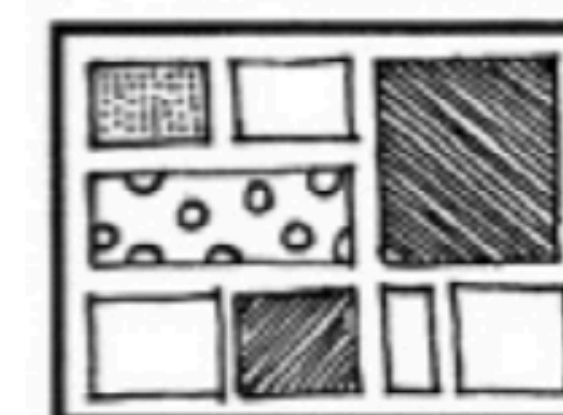
Annotated Chart



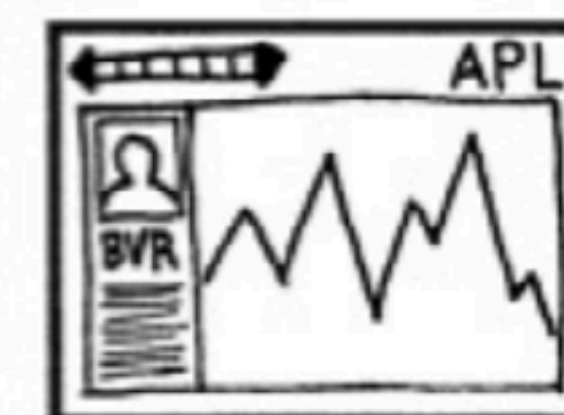
Partitioned Poster



Flow Chart



Comic Strip



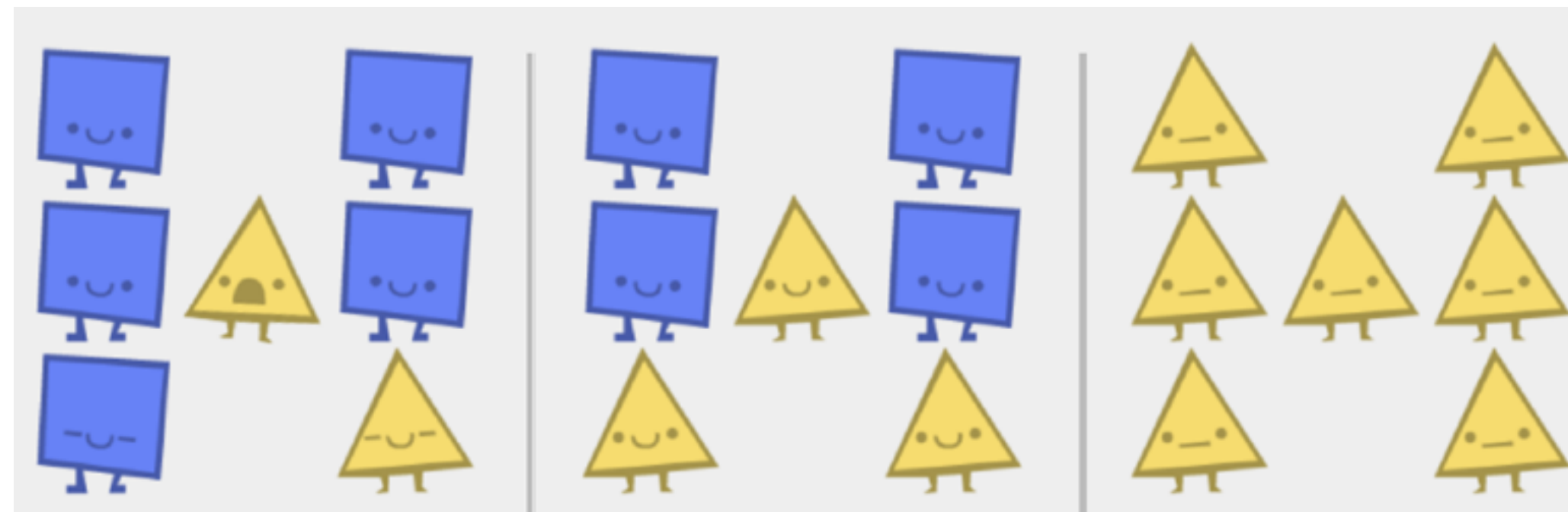
Slide Show



Film/Video/Animation

Activity

Parable of Polygons



<https://ncase.me/polygons/>

What techniques?

How are they directing attention?

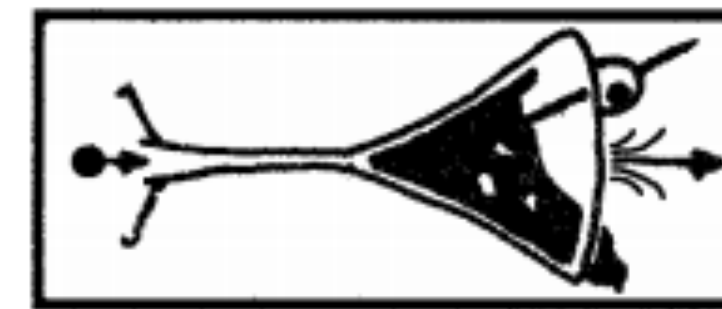
Is it effective / engaging?



Montage / editing



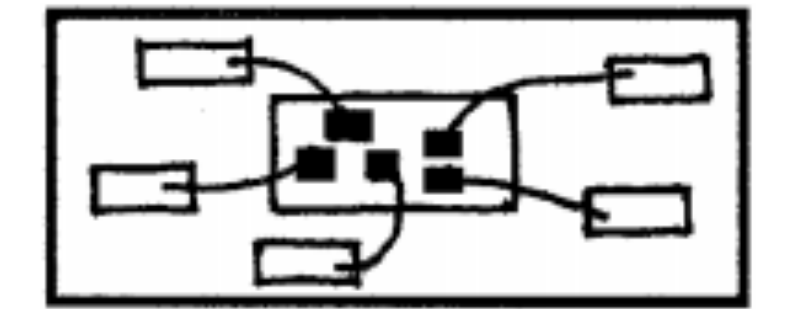
Ensemble staging / direction



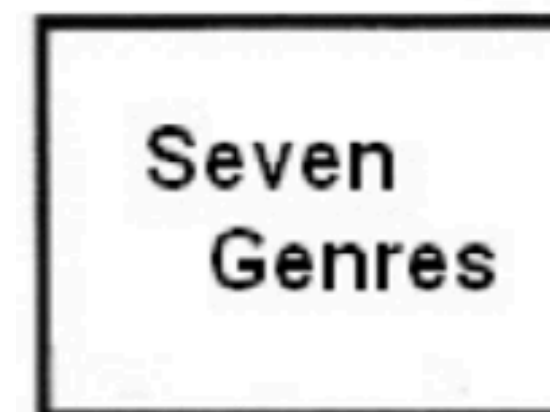
Martini glass



Interactive slideshow



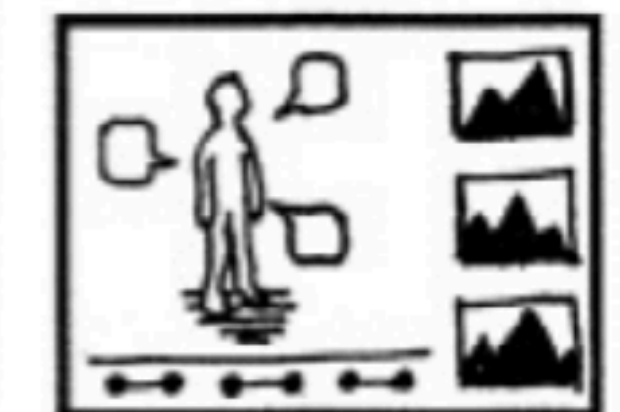
Drill-down story



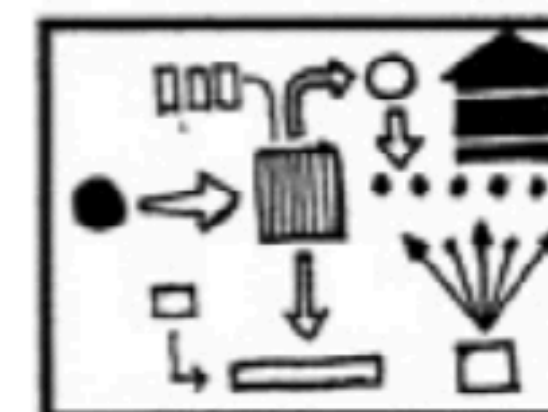
Magazine Style



Annotated Chart



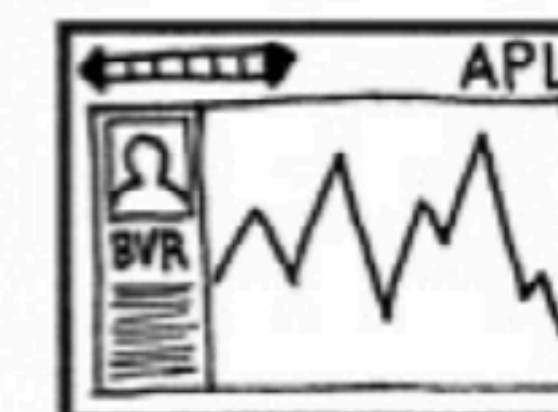
Partitioned Poster



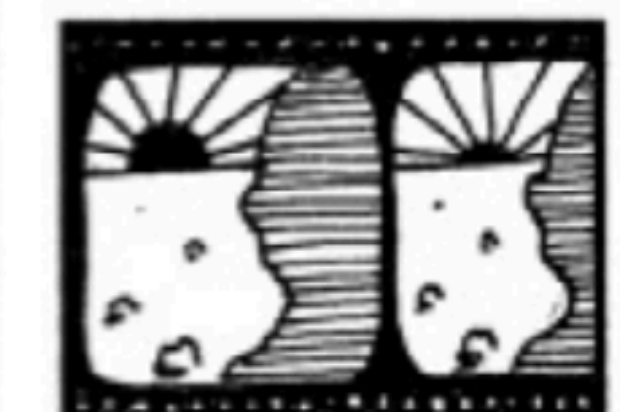
Flow Chart



Comic Strip

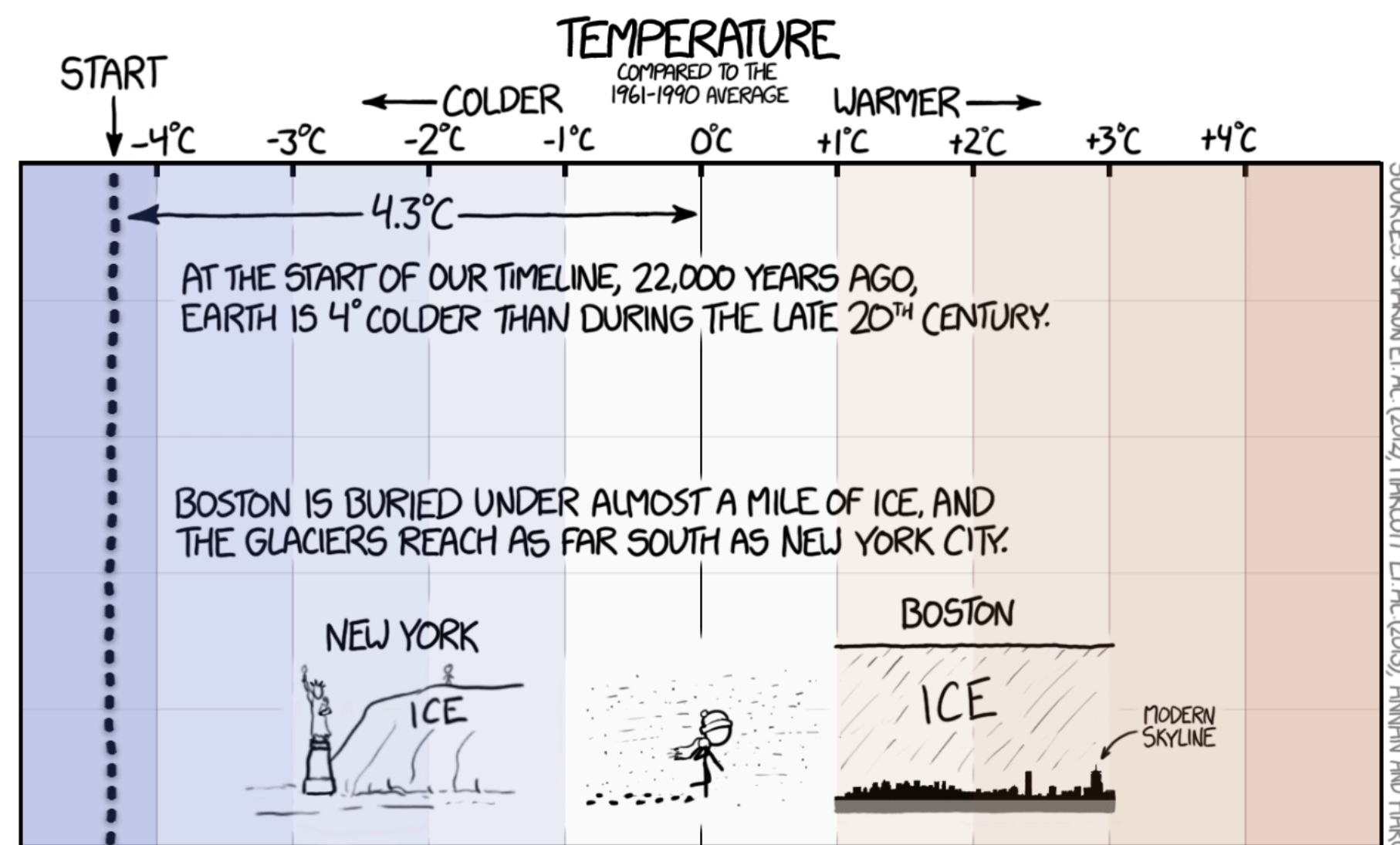


Slide Show



Film/Video/Animation

Activity



<https://xkcd.com/1732/>

What techniques?

How are they directing attention?

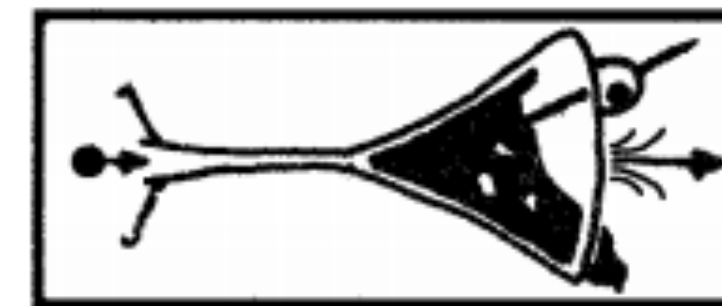
Is it effective / engaging?



Montage / editing



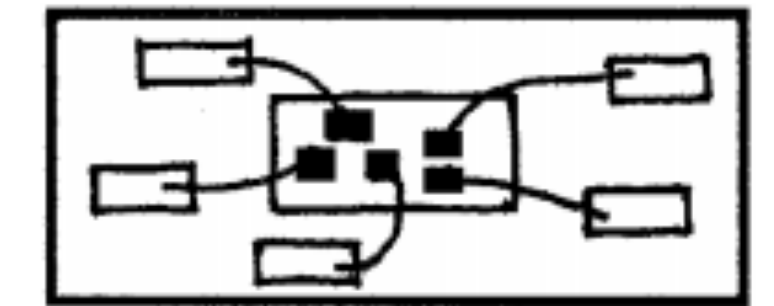
Ensemble staging / direction



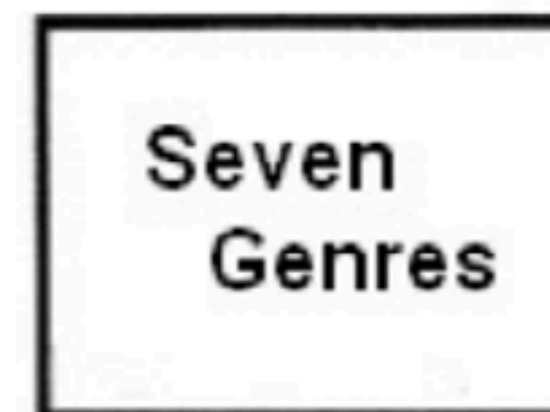
Martini glass



Interactive slideshow



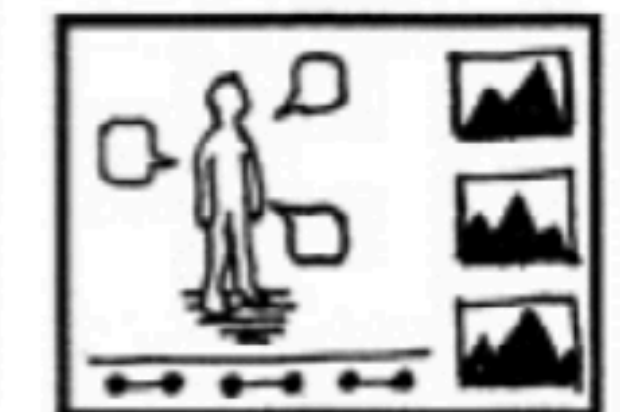
Drill-down story



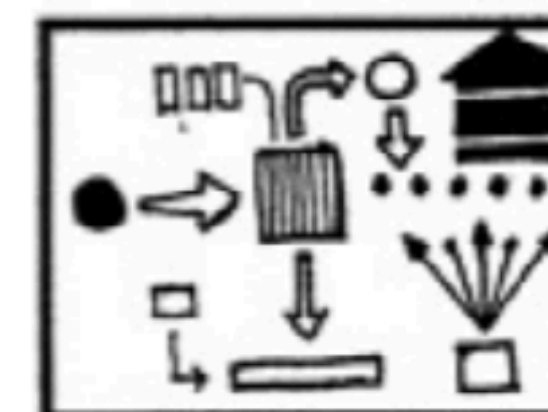
Magazine Style



Annotated Chart



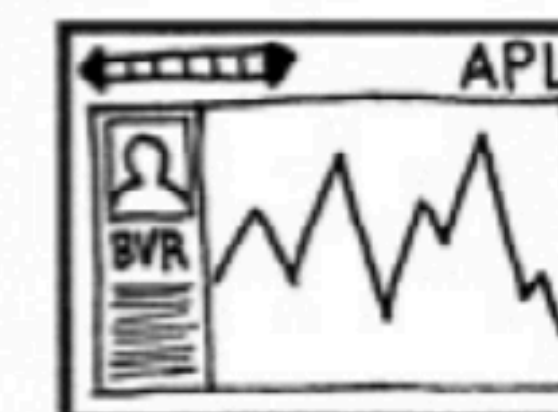
Partitioned Poster



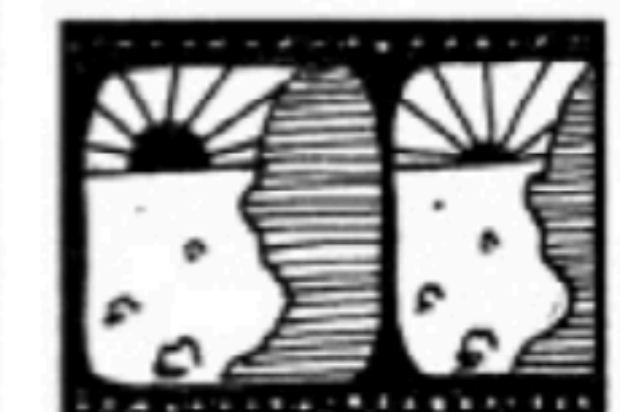
Flow Chart



Comic Strip



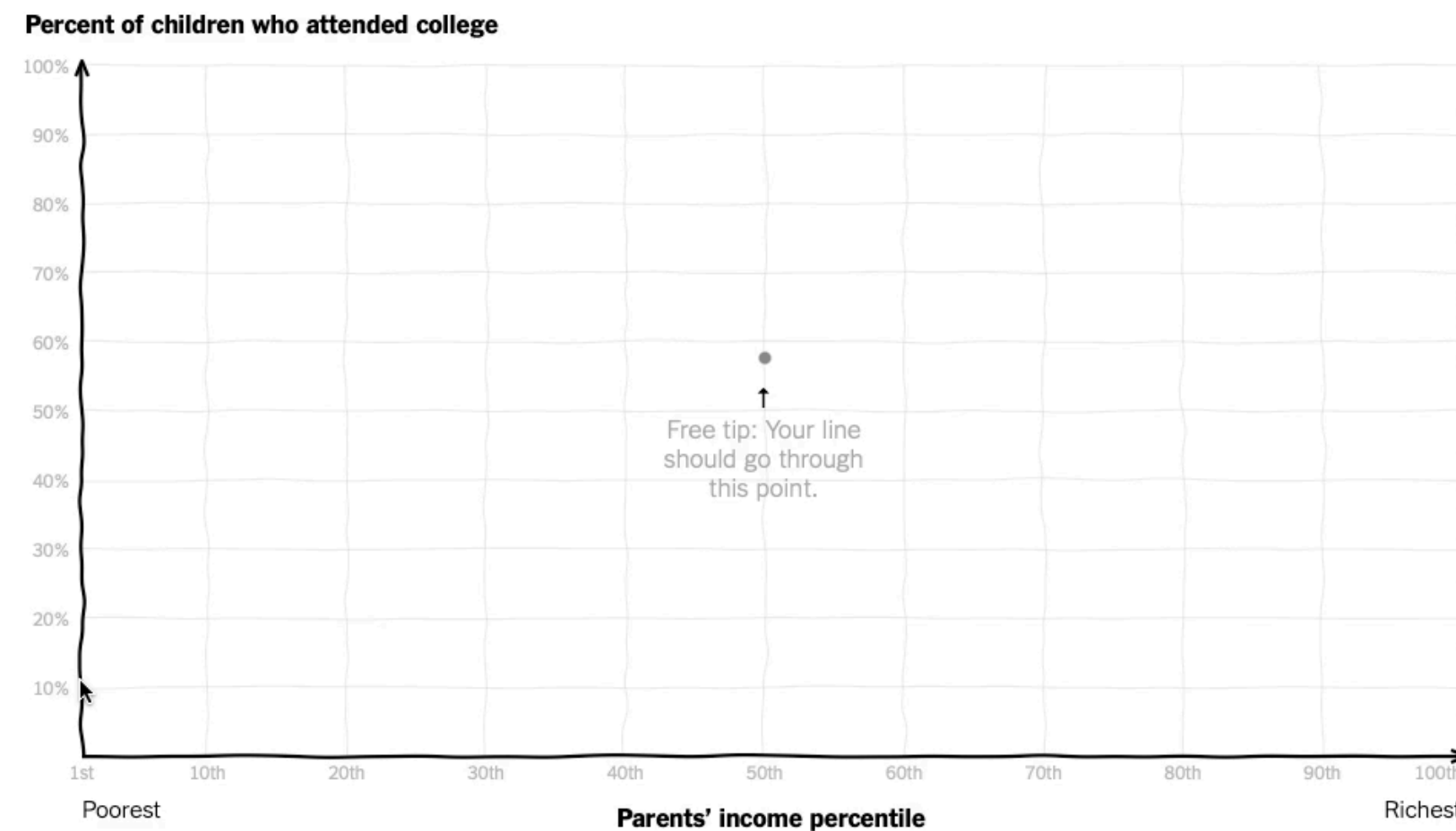
Slide Show



Film/Video/Animation

Activity

Draw your line on the chart below



<https://www.nytimes.com/interactive/2015/05/28/upshot/you-draw-it-how-family-income-affects-childrens-college-chances.html>

What techniques?

How are they directing attention?

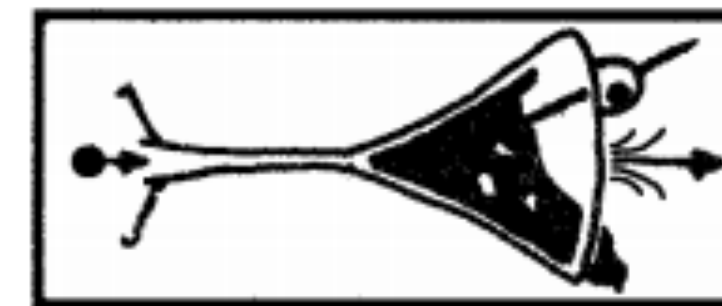
Is it effective / engaging?



Montage / editing



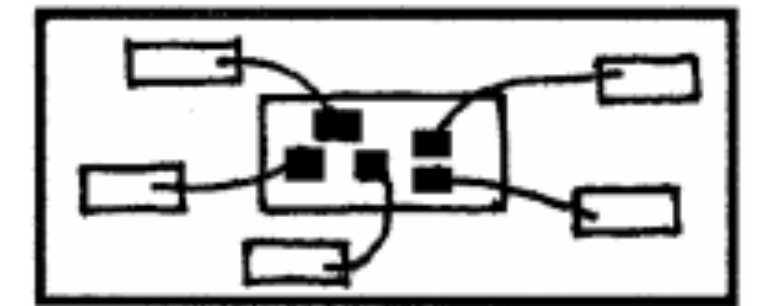
Ensemble staging / direction



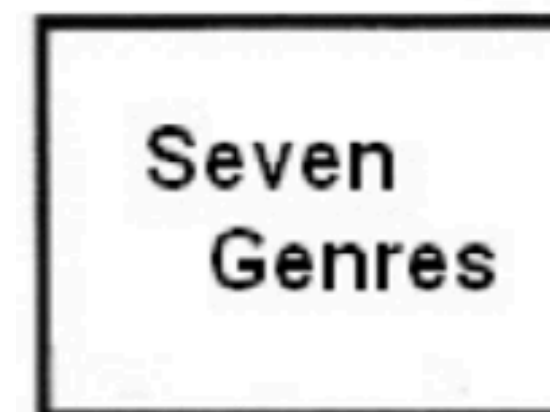
Martini glass



Interactive slideshow



Drill-down story



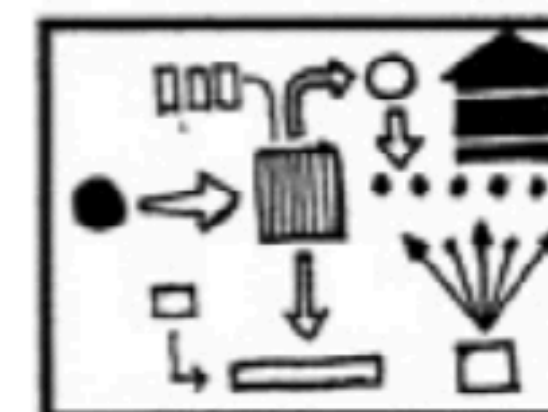
Magazine Style



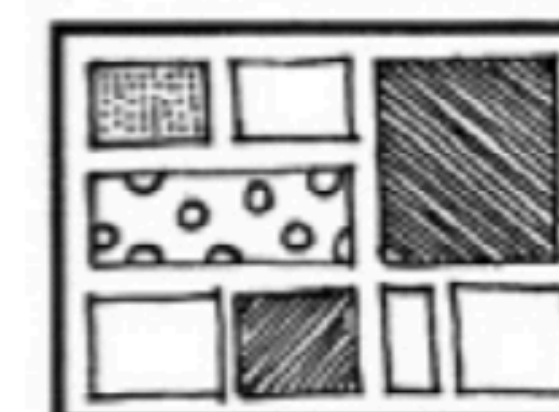
Annotated Chart



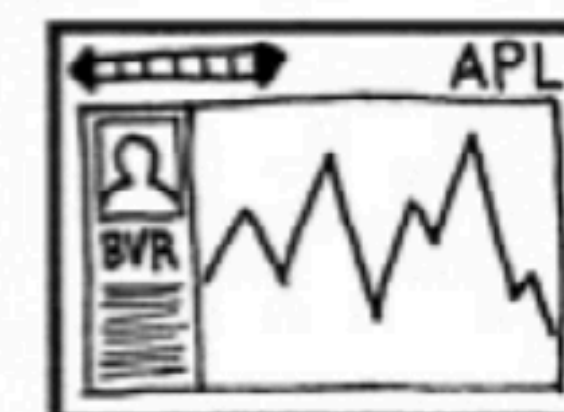
Partitioned Poster



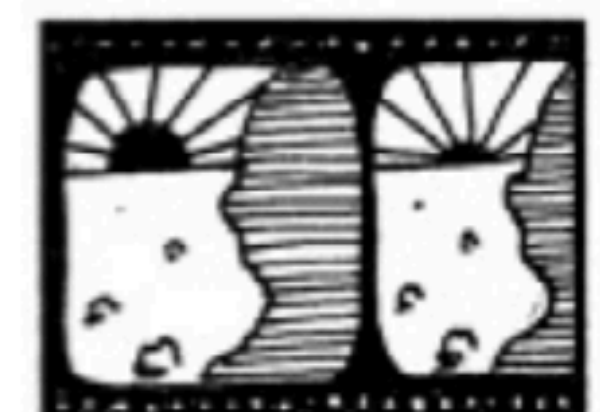
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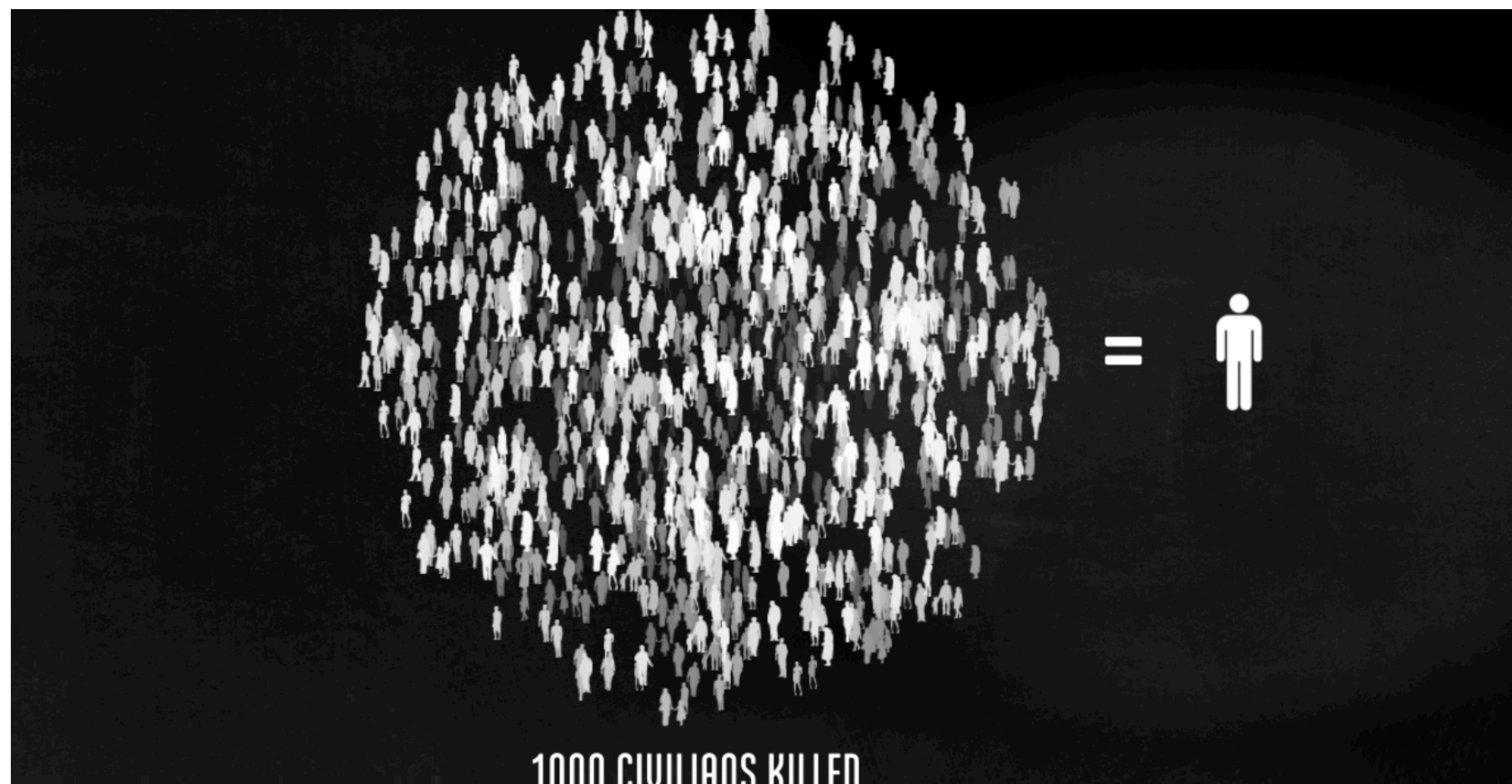
Slide Show



Film/Video/Animation

Activity

The Fallen of World War II



<http://www.fallen.io/ww2/>

What techniques?

How are they directing attention?

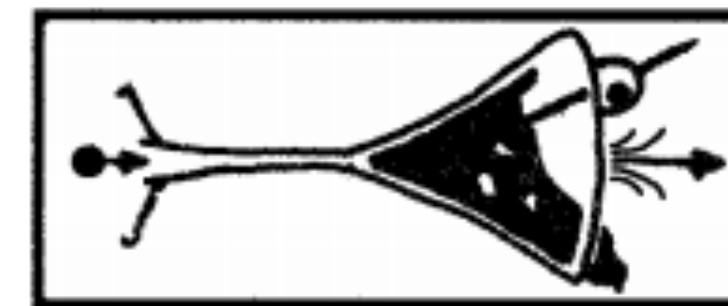
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Montage / editing



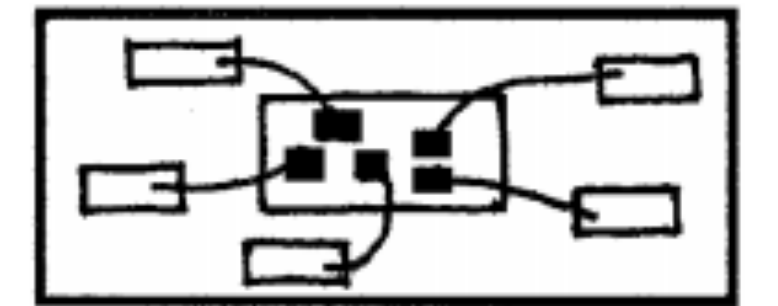
Ensemble staging / direction



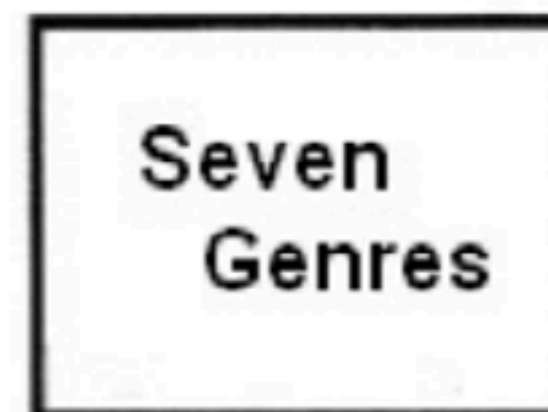
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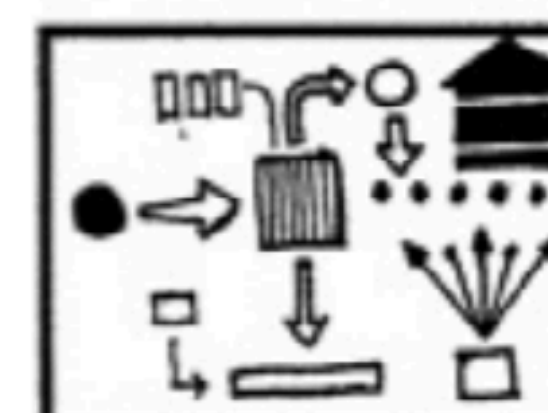
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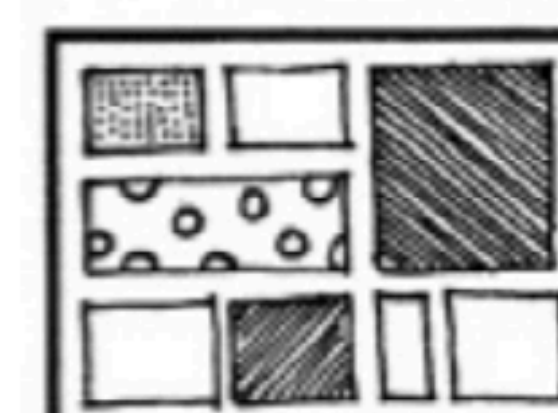
Annotated Chart



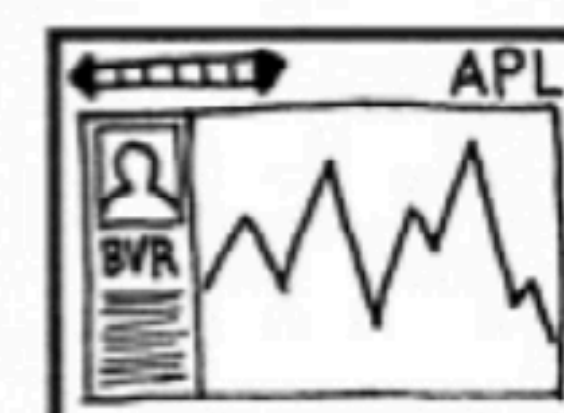
Partitioned Poster



Flow Chart



Comic Strip

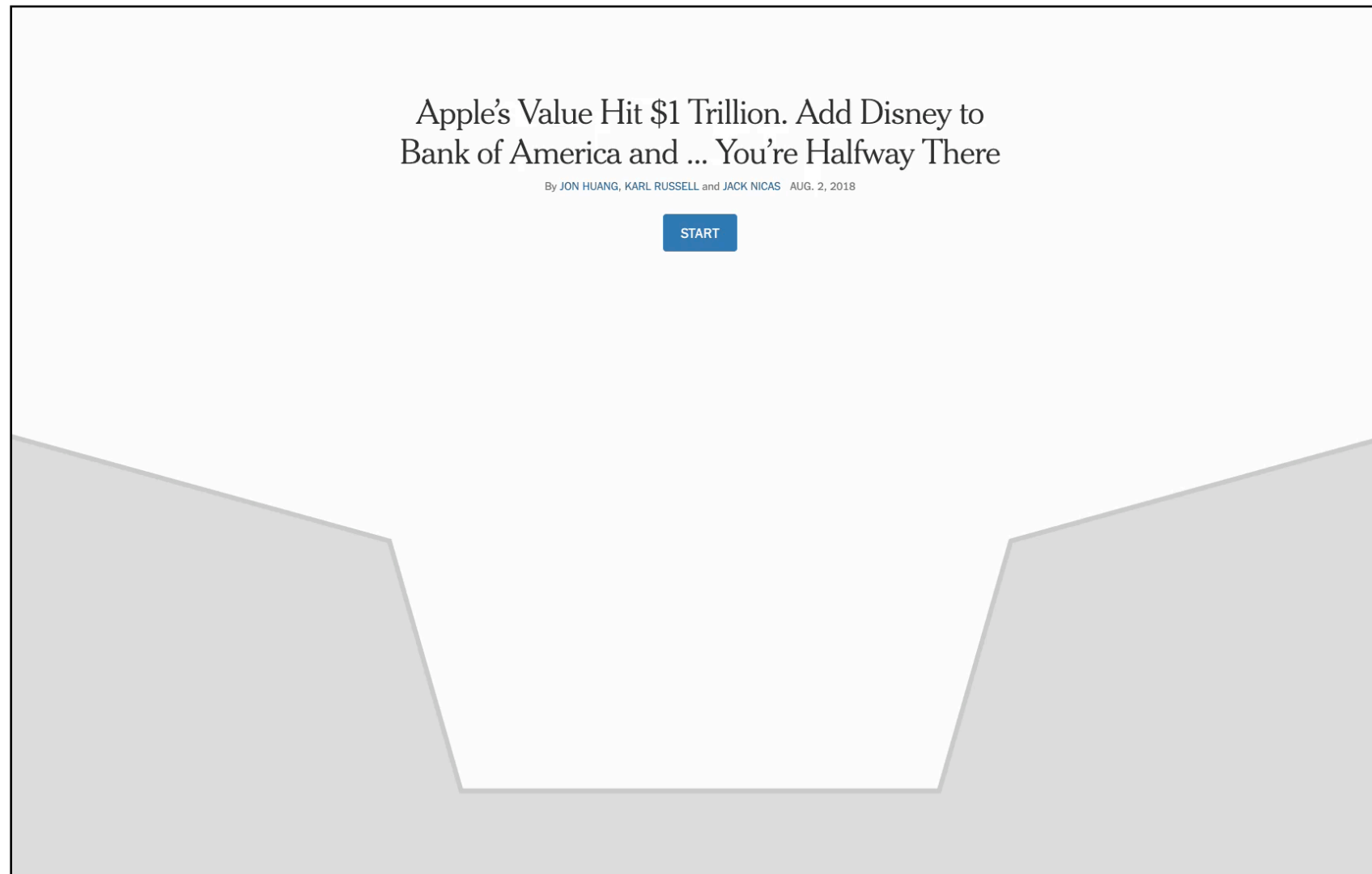


Slide Show



Film/Video/Animation

Activity



<https://www.nytimes.com/interactive/2018/08/02/technology/apple-trillion-market-cap.html>

What techniques?

How are they directing attention?

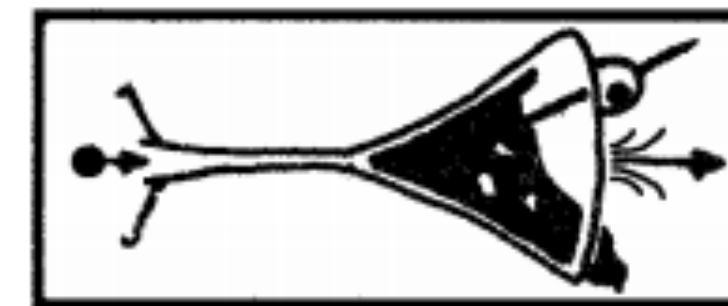
Is it effective / engaging?



Montage / editing



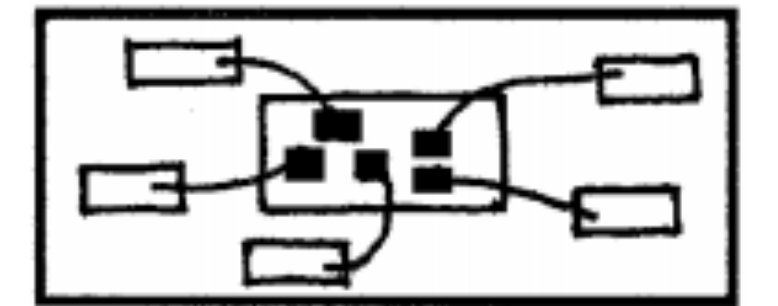
Ensemble staging / direction



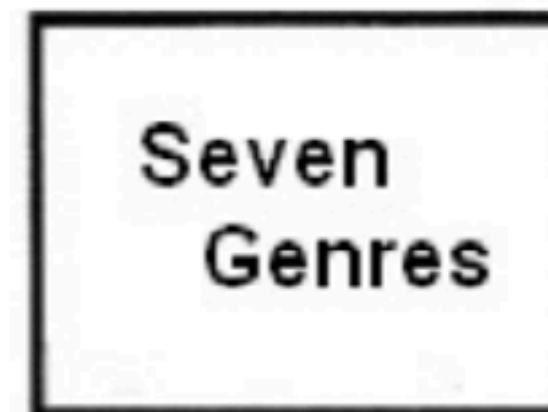
Martini glass



Interactive slideshow



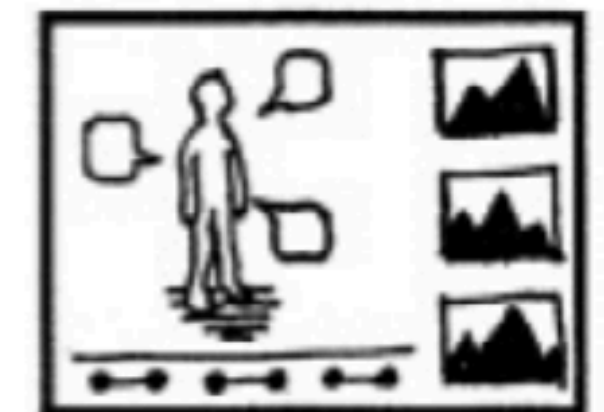
Drill-down story



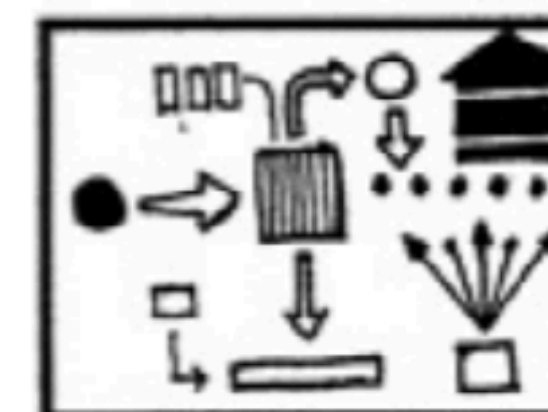
Magazine Style



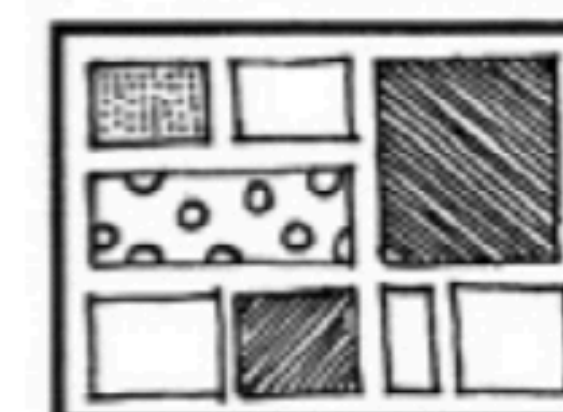
Annotated Chart



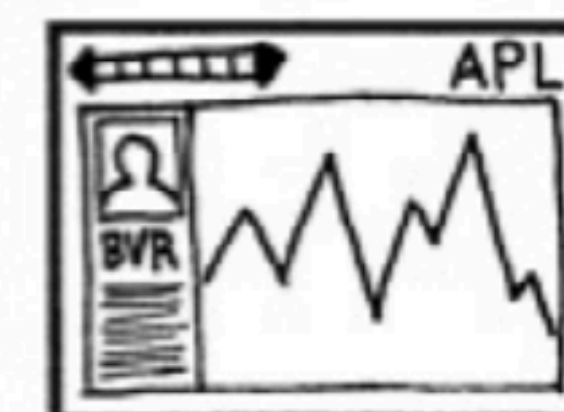
Partitioned Poster



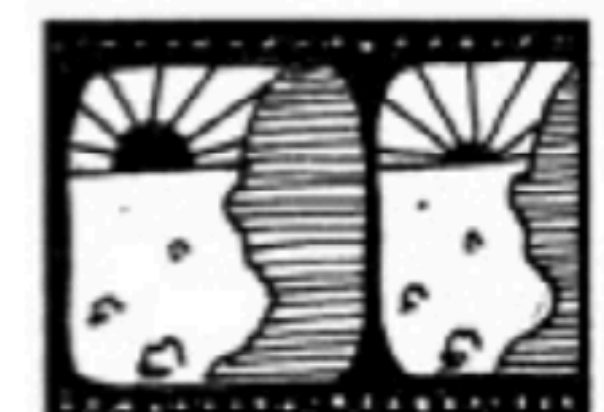
Flow Chart



Comic Strip



Slide Show



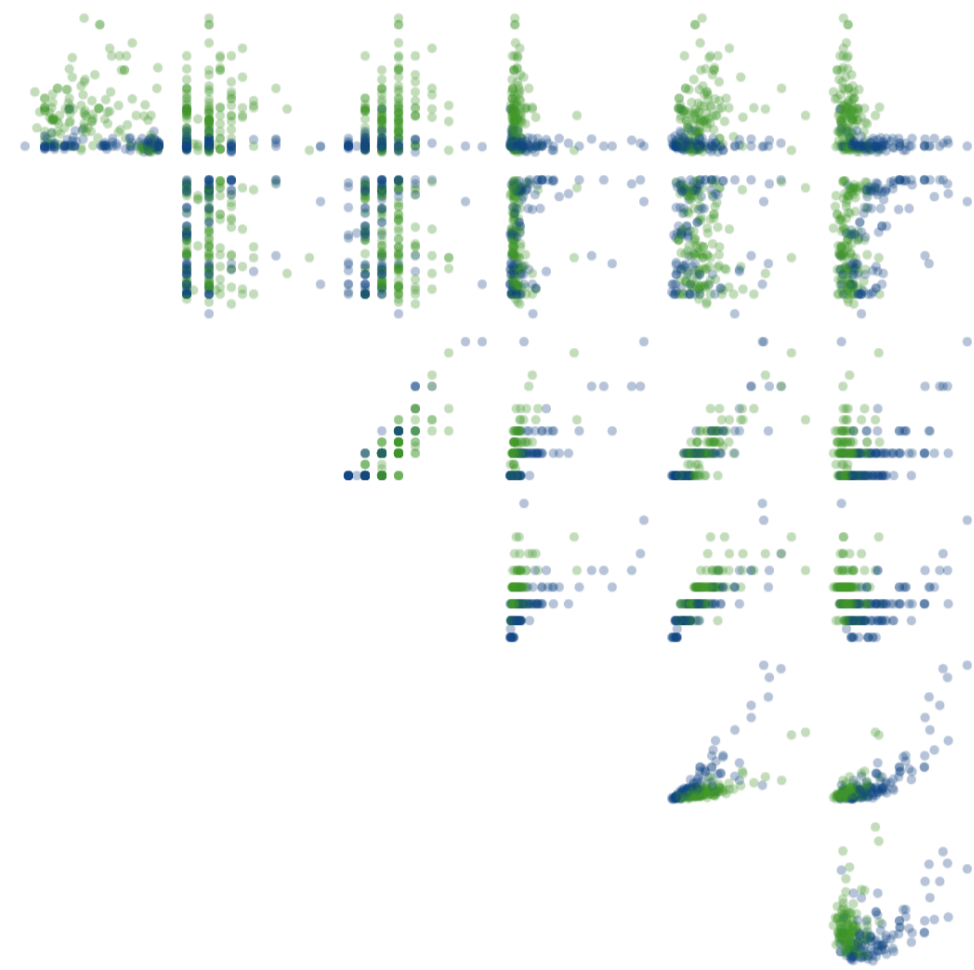
Film/Video/Animation

Activity

A visual introduction to machine learning

English

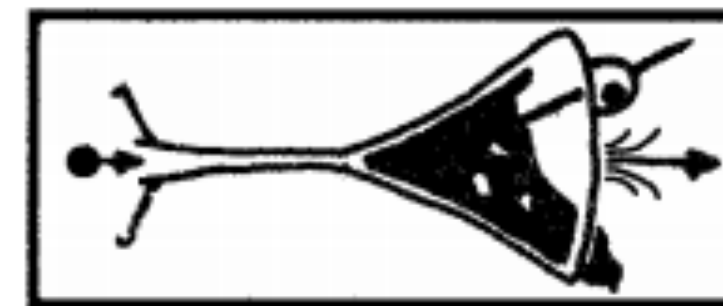
In machine learning, computers apply **statistical learning** techniques to automatically identify patterns in data. These techniques can be used to make highly accurate predictions.



Montage / editing



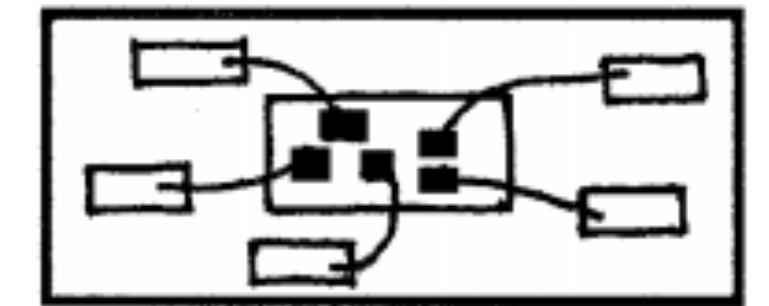
Ensemble staging / direction



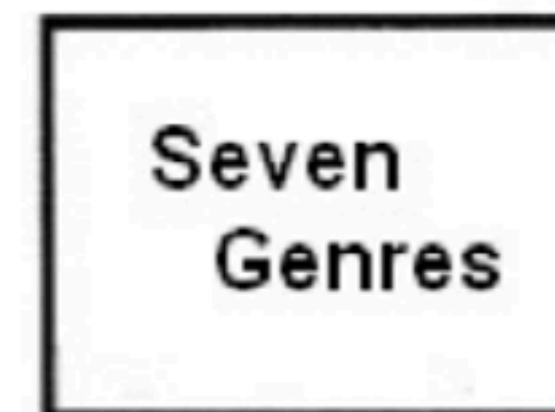
Martini glass



Interactive slideshow



Drill-down story



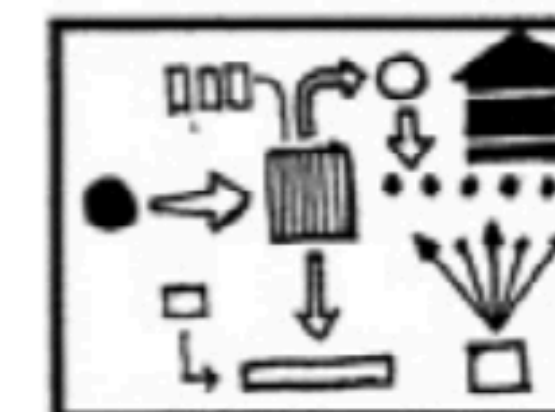
Magazine Style



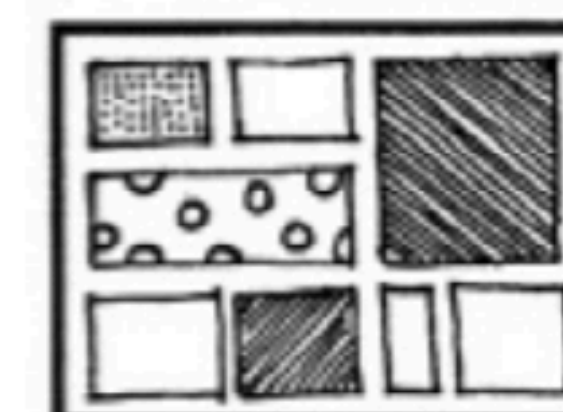
Annotated Chart



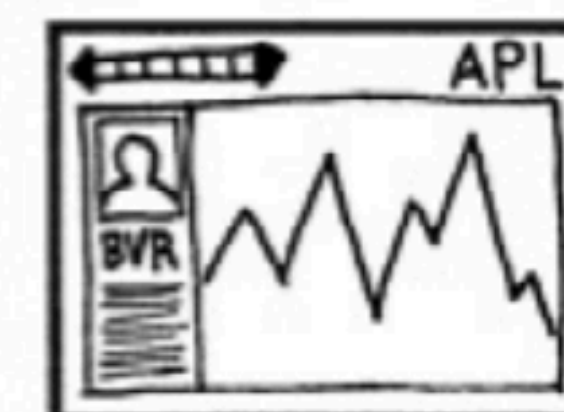
Partitioned Poster



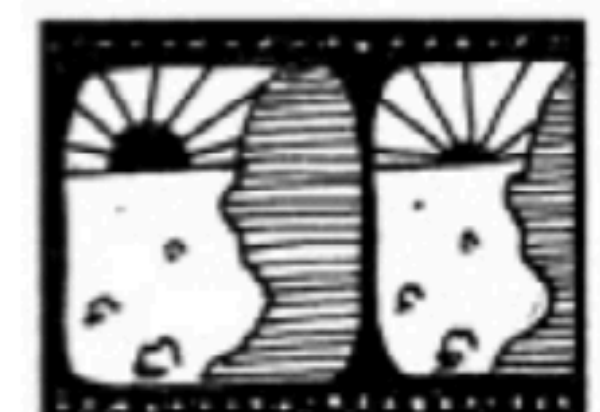
Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

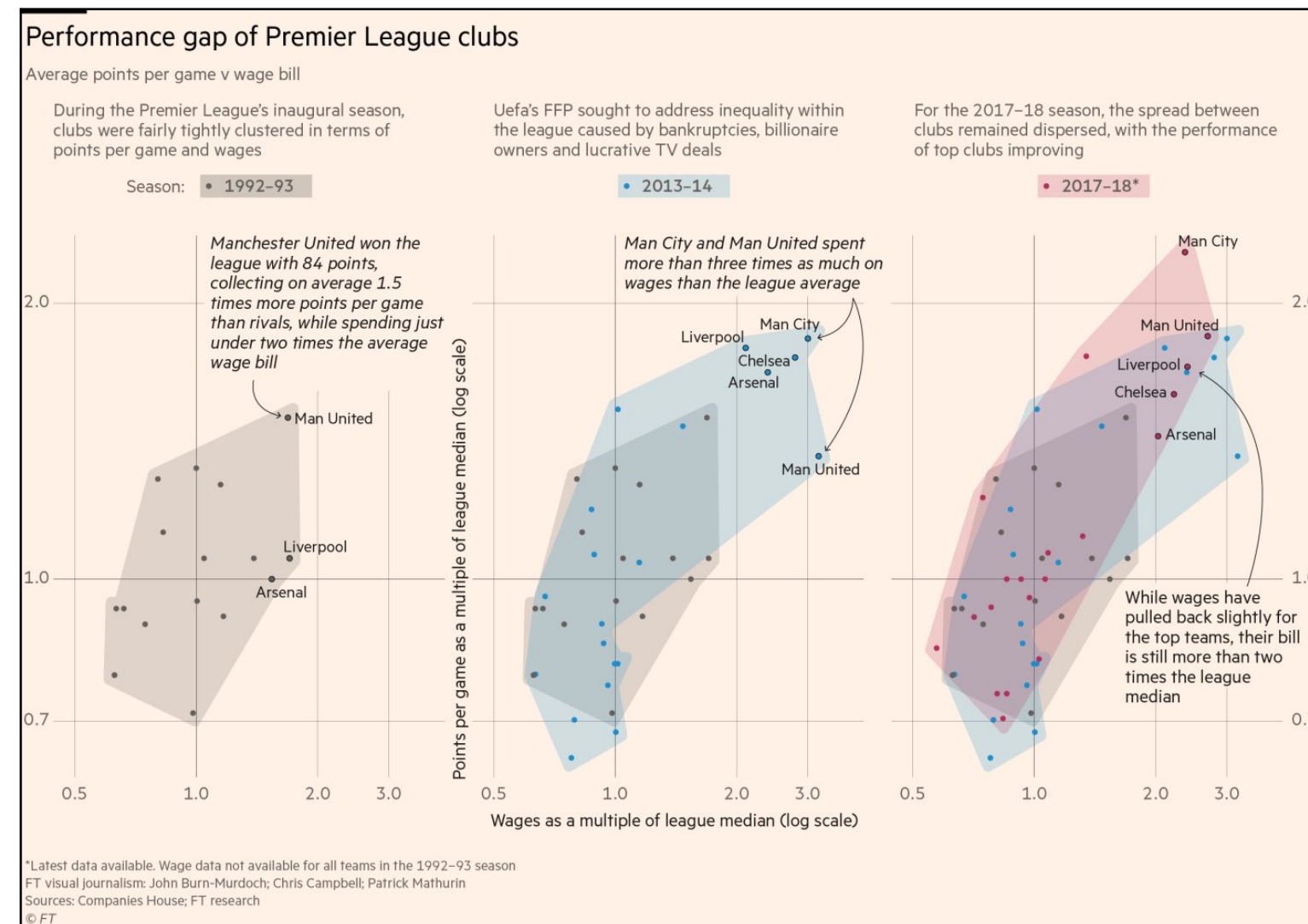
What techniques?

How are they directing attention?

Is it effective / engaging?

<https://www.r2d3.us/visual-intro-to-machine-learning-part-1/>

Activity



<https://x.com/ftdata/status/1231157648839651328/photo/1>

What techniques?

How are they directing attention?

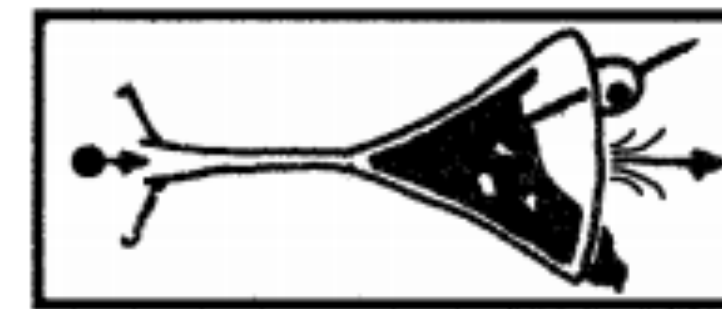
Is it effective / engaging?



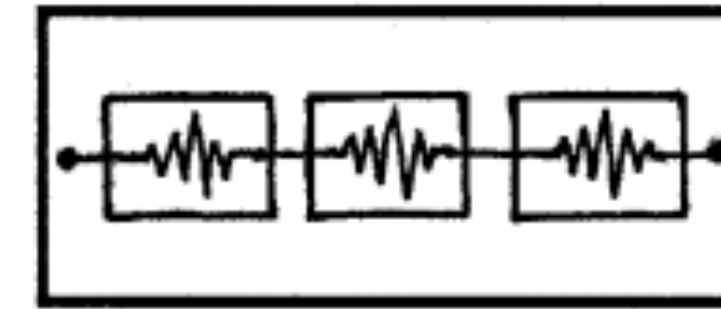
Montage / editing



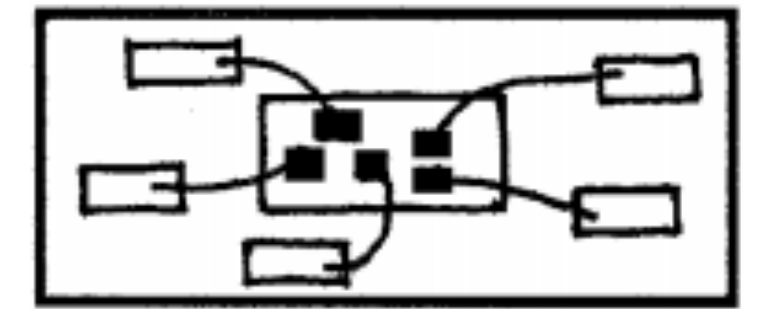
Ensemble staging / direction



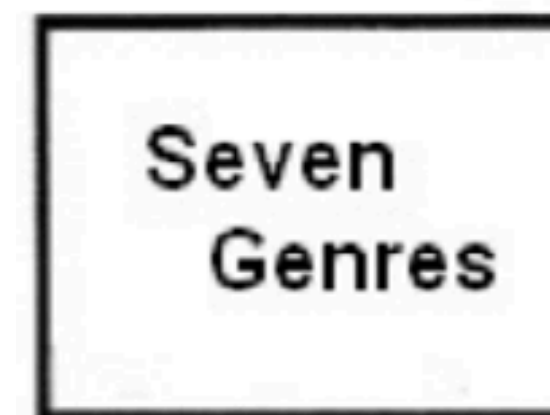
Martini glass



Interactive slideshow



Drill-down story



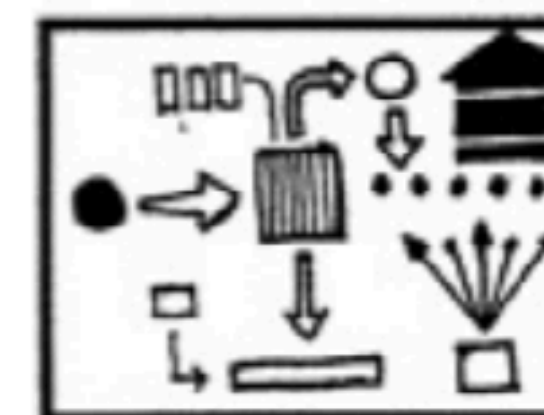
Magazine Style



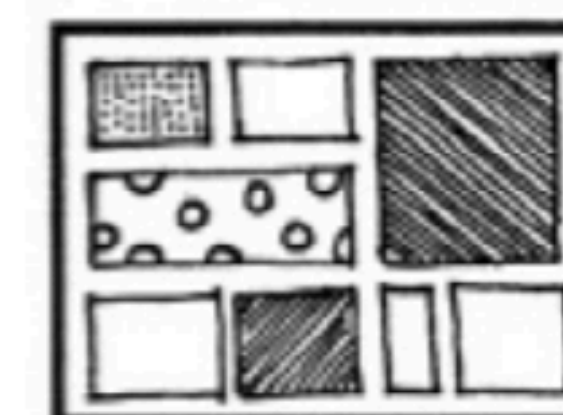
Annotated Chart



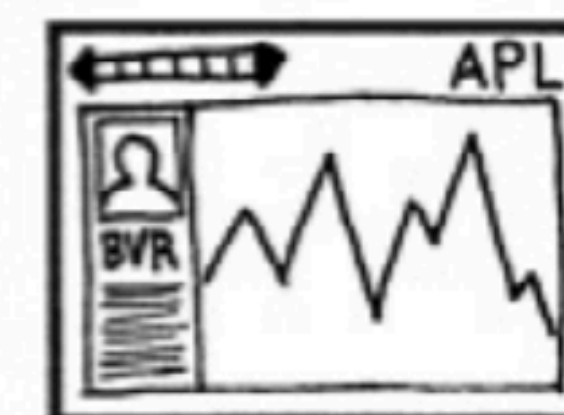
Partitioned Poster



Flow Chart



Comic Strip

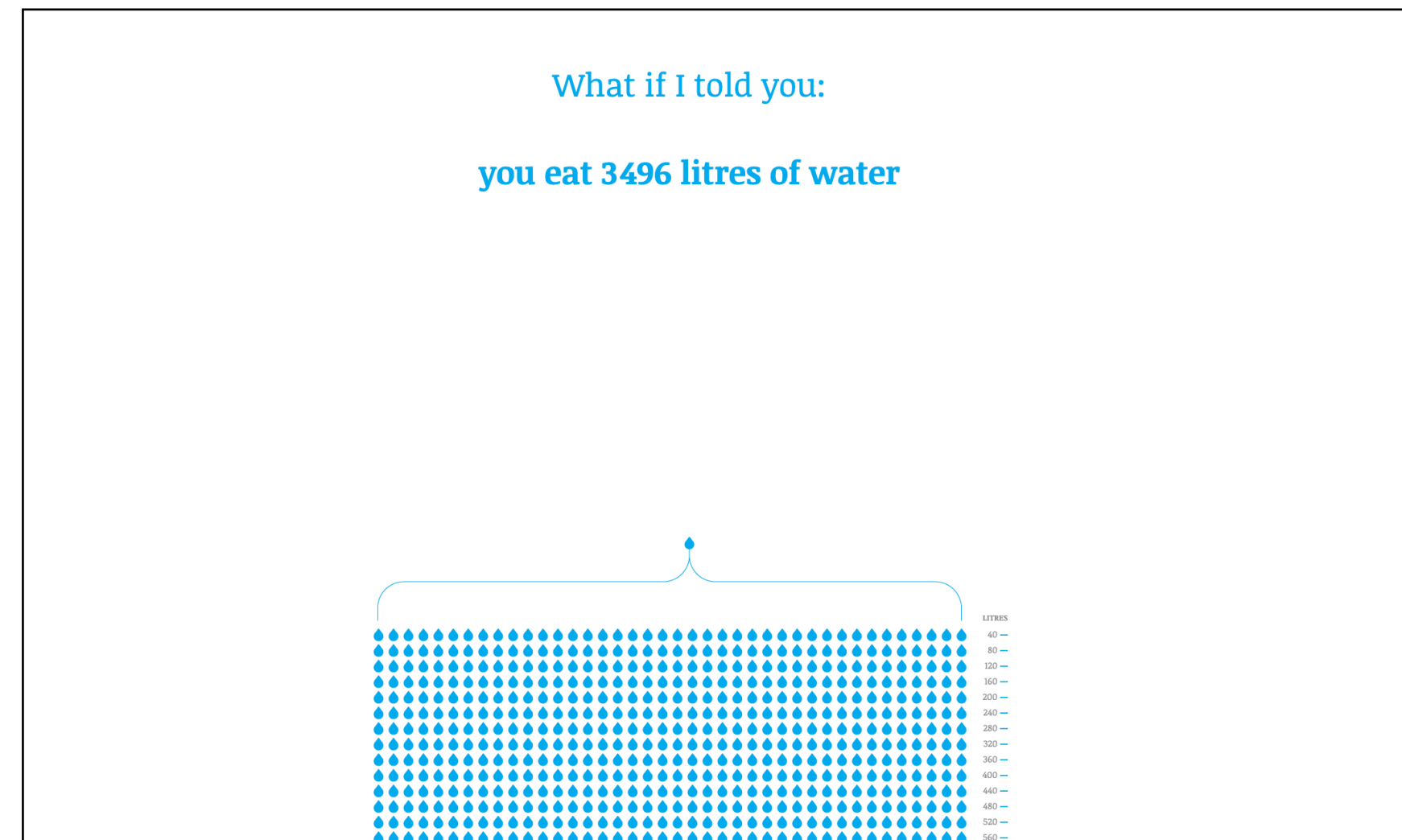


Slide Show



Film/Video/Animation

Activity



<https://thewaterweeat.com/>

What techniques?

How are they directing attention?

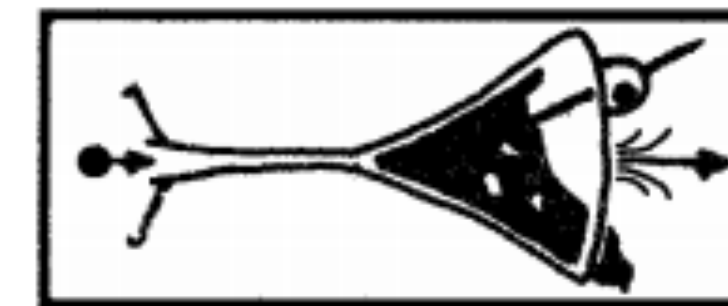
Is it effective / engaging?



Montage / editing



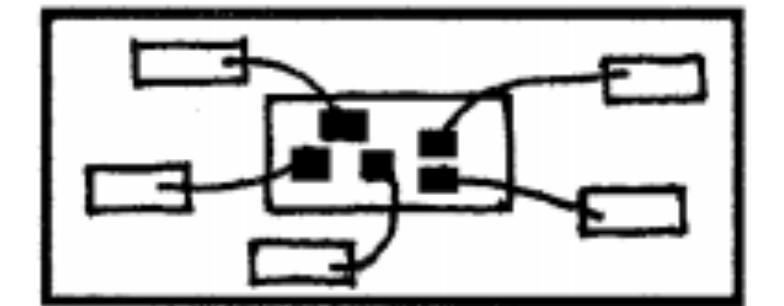
Ensemble staging / direction



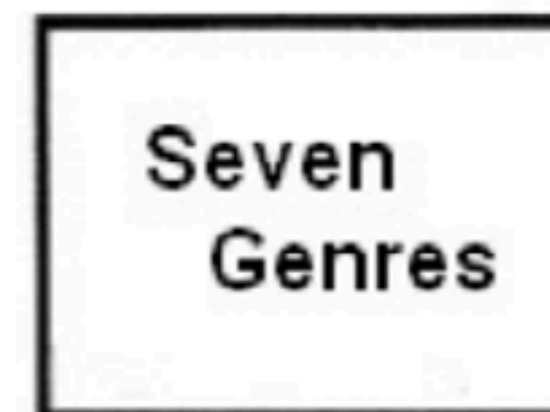
Martini glass



Interactive slideshow



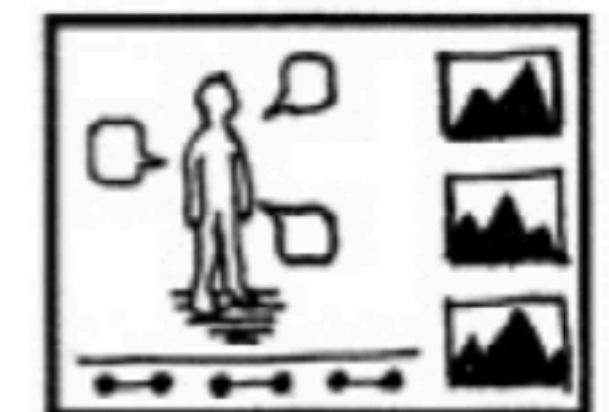
Drill-down story



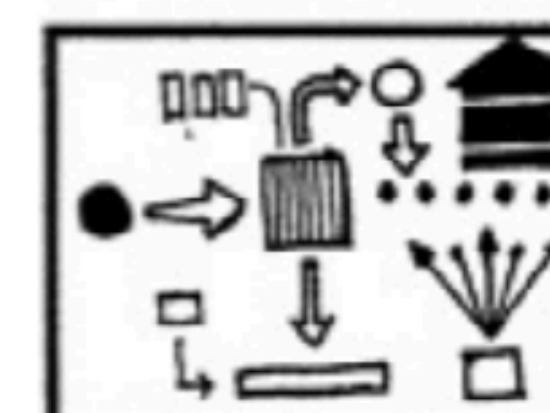
Magazine Style



Annotated Chart



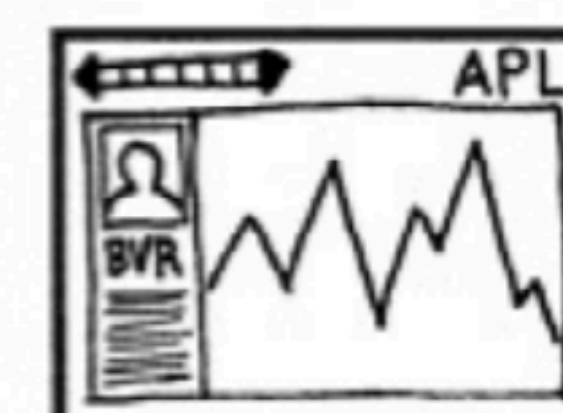
Partitioned Poster



Flow Chart



Comic Strip

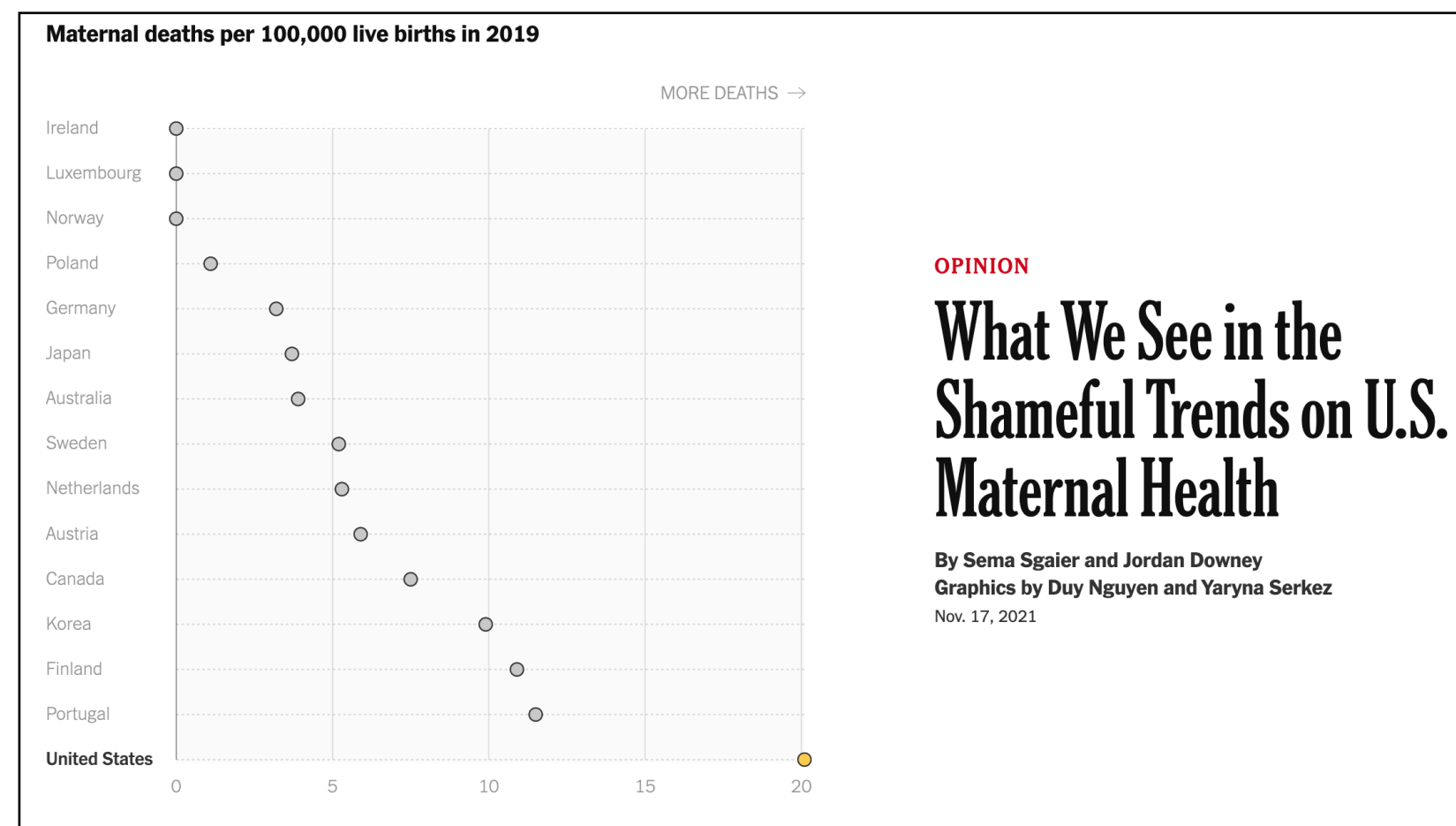


Slide Show



Film/Video/Animation

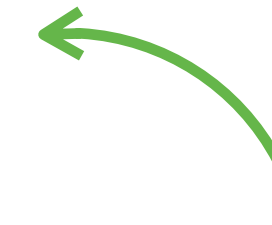
Other examples



<https://www.nytimes.com/interactive/2021/11/17/opinion/maternal-pregnancy-health.html>

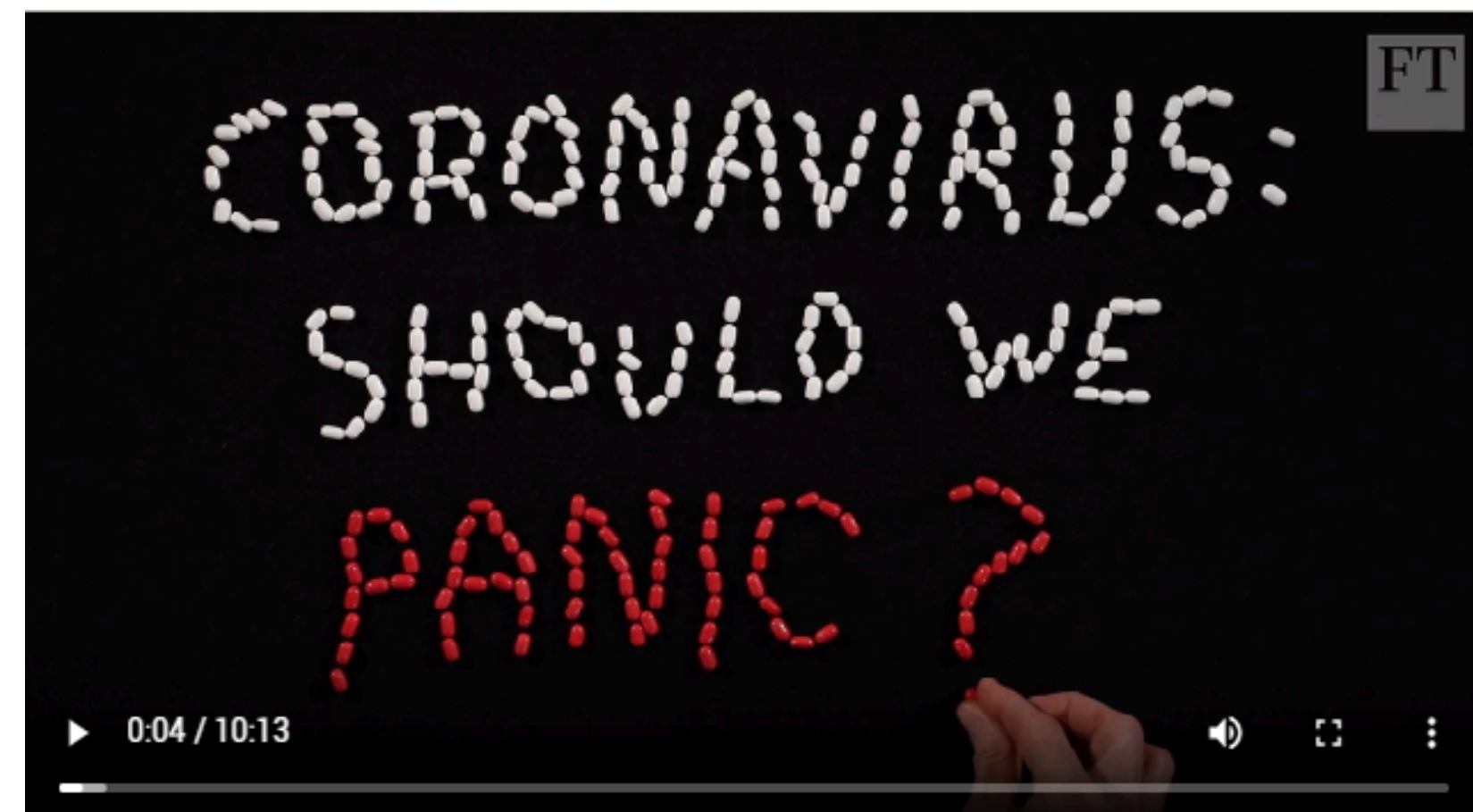


<https://www.nytimes.com/interactive/2020/03/22/world/coronavirus-spread.html>



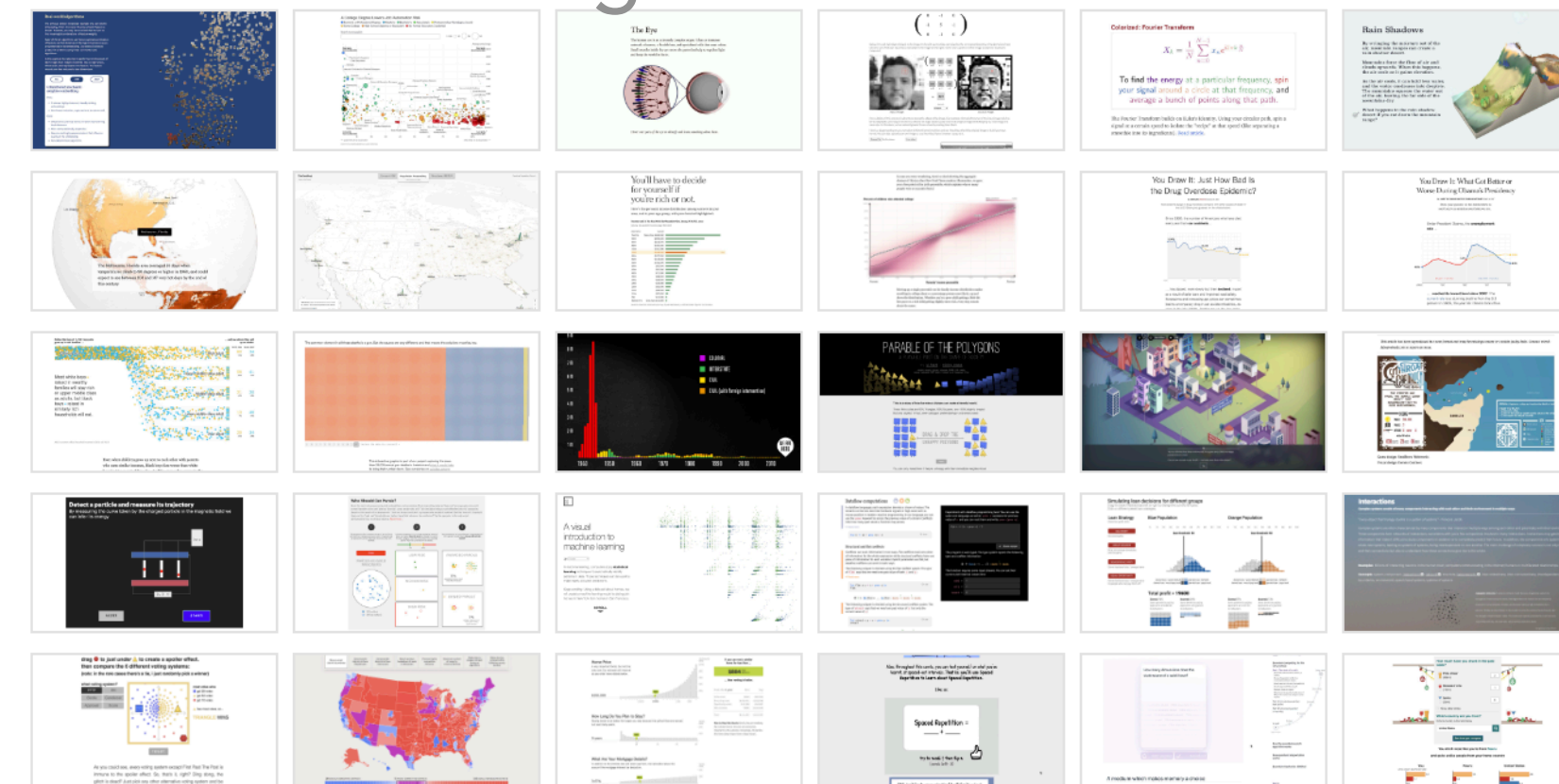
N.B.: This article uses excellent vis and interactivity, but its content is controversial

P.S. free access to NYT via UMD
<https://lib.guides.umd.edu/UMDNYTAccessPass>



<https://tinyurl.com/tbesbd4>

Communicating with Interactive Articles



<https://distill.pub/2020/communicating-with-interactive-articles/>

Affordances of the format

Connecting people and data

Making systems playful

Promoting self reflection

Personalizing reading

Use text and annotation!

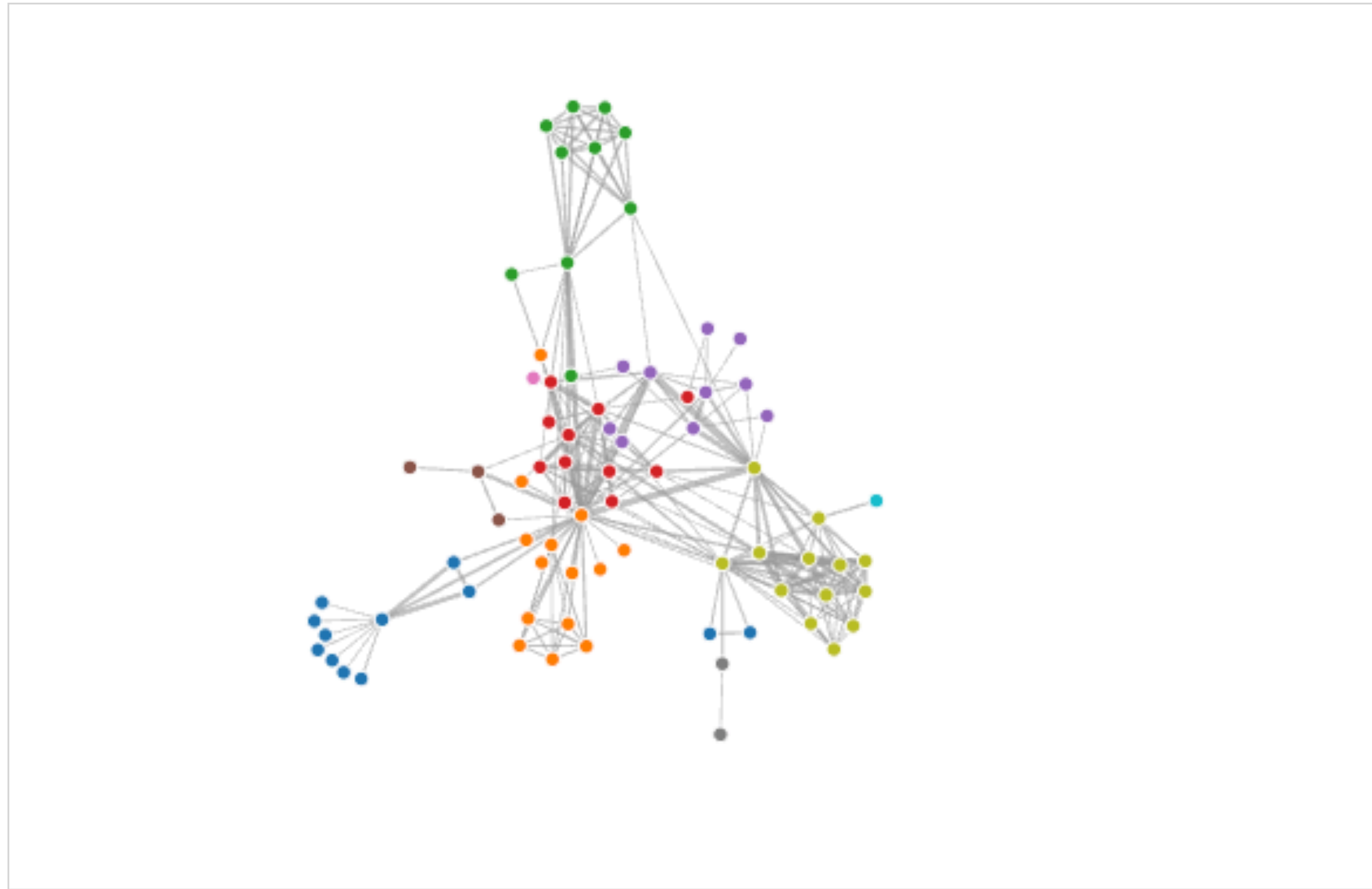
Think about **order** and how **explicit** it needs to be

Tension, suspense, surprise, uncertainty...

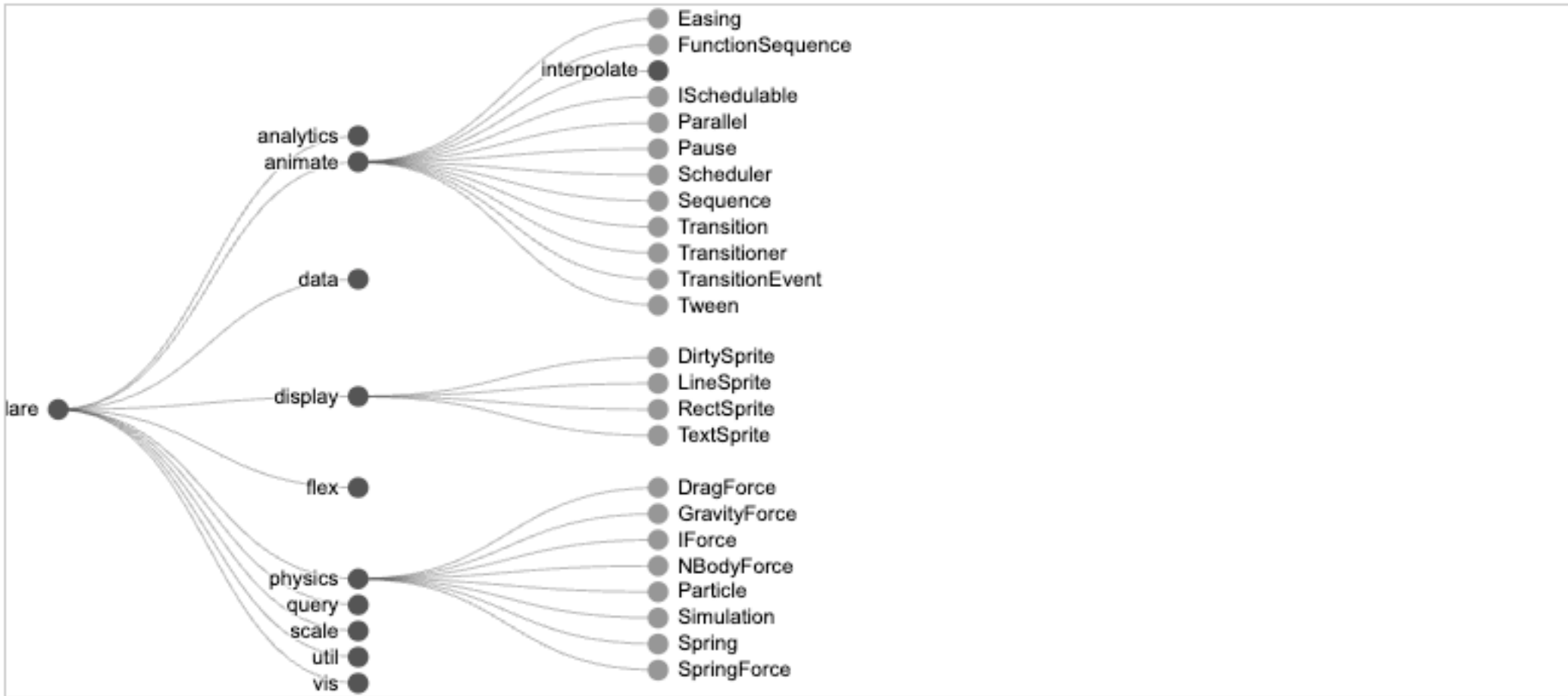
Take inspiration from other work, like film, comic, art

*I definitely encourage
to make an engaging
storytelling vis/video
for your final project!*

Wed Lab



Force-directed layout



Tree visualization